The target-free clique conjecture for threshold-linear networks: complete exact maps for $n \leq 7$, the $\varepsilon \to 0$ map at $n = 8$, infinite families, and unbounded thresholds

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Abstract

For combinatorial threshold-linear networks (CTLNs), Curto and Morrison asked whether the supports of stable fixed points are exactly the target-free cliques of the underlying digraph. Geneson disproved this with a six-node graph near $q = 1$, where $q = \delta(1-\varepsilon)/\varepsilon > 1$ is the legality parameter, and proved that the conjecture holds once $q \geq n - 2 - (n-3)\varepsilon/2$. We study the gap between these two results with exact arithmetic throughout. First, a reduction theorem shows that at every legal parameter the conjecture on n nodes is equivalent to the statement that no non-complete digraph on $k \leq n$ nodes has a positive, linearly stable full-support fixed point. Second, using the rank-one structure of the weight matrix, an exact $\varepsilon \to 0$ spectral criterion, Hurwitz determinants with a three-valued treatment of boundary points, a cylindrical decomposition of the two-parameter legal region, and a ray reduction to the legality curve, we obtain complete certified maps of stable non-clique full supports: there are none for $n \leq 5$; at $n = 6$ exactly three of the 1,540,944 digraphs have one; at $n = 7$ exactly 124 of the 882,033,440 digraphs do, one of which is invisible in the $\varepsilon \to 0$ limit. The exact thresholds are $q_6^* \approx 1.1664$, a root of an explicit sextic, and $q_7^* = \rho \approx 1.3247$, the plastic number: for $n \leq 7$ the conjecture holds at every legal parameter with $q \geq \rho$ and fails for every $q \in (1, \rho)$. Third, at $n = 8$ we certify the complete $\varepsilon \to 0$ limit map: of the 801,558,826,511 digraphs not excluded by a Lean-checked edge bound, exactly 9,133 non-complete graphs have a positive, limit-stable full support, 9,072 of them not closed-twin lifts of seven-node counterexamples, and the limit threshold $q_8^*(\infty) \approx 1.7107$ is a root of an explicit degree-15 polynomial, so $q_8^* > q_7^*$; this list was verified by two independent machine campaigns but, unlike the $n \leq 7$ lists, not by hand, and the finite-ε map at $n = 8$ remains open. Fourth, closed-twin blow-ups yield infinite families: counterexamples on every $n \geq 6$ nodes, finite-ε-only counterexamples on every $n \geq 7$ nodes, a proof that the critical value q_n^* tends to infinity, and a single legal parameter point at which stable non-clique supports of every size $n \geq 6$ exist. Several of the necessary conditions used to prune the search are machine-checked in Lean 4.

1 Introduction

1.1 The conjecture

A *threshold-linear network* on n neurons is the dynamical system

$$\frac{dx_i}{dt} = -x_i + \Big[\sum_{j=1}^{n} W_{ij} x_j + \theta\Big]_+, \qquad i = 1, \ldots, n, \tag{1}$$

with $[u]_+ = \max(u, 0)$ and a constant external drive $\theta > 0$. A *combinatorial* TLN (CTLN) builds W from a directed graph G and two parameters $\varepsilon, \delta > 0$: weak inhibition $-1 + \varepsilon$ along edges and strong inhibition $-1 - \delta$ along non-edges (Section 2). The CTLN is *legal* when $\delta > 0$, $0 < \varepsilon < \delta/(\delta + 1)$,

which we write as $q := \delta(1-\varepsilon)/\varepsilon > 1$. Fixed points of (1) are indexed by their supports $\sigma \subseteq [n]$, and a central theme of the CTLN programme is to read the fixed-point structure off the graph [3].

A subset σ is a *clique* if every pair of distinct nodes of σ is joined by edges in both directions; a node $k \notin \sigma$ is a *target* of σ if every node of σ sends an edge to k; σ is *target-free* if it is a clique with no target. Open Question 10 of Curto and Morrison [3] is the *target-free clique conjecture*: *the supports of stable fixed points of a CTLN are exactly its target-free cliques.* One direction is a classical graph rule: every target-free clique supports a stable fixed point at every legal parameter (Lemma 2.5). The content of the conjecture is the converse.

1.2 Geneson's counterexample and bound

Geneson [4] disproved the conjecture with an explicit graph on six nodes and $\delta = \frac{29}{25}\varepsilon$ for all sufficiently small ε (so $q \approx 29/25$), and proved that it does hold whenever

$$q \;\geq\; n - 2 - \tfrac{(n-3)\varepsilon}{2}. \tag{2}$$

This leaves a gap between q slightly above 1 and q near $n-2$. The present paper maps that gap exactly for $n \leq 7$, maps its $\varepsilon \to 0$ limit at $n = 8$, and studies its behaviour for large n.

1.3 Our results

Throughout, results are labelled PROVED (a proof is given here or in the project notes), CERTIFIED (an exact rational or algebraic computation recorded in a re-runnable certificate file), or LEAN (machine-checked in Lean 4 with Mathlib; marked ᴸ in statements). No floating-point number enters any decision; decimals in the text are display approximations of exactly specified algebraic numbers. Graphs on $n \leq 7$ nodes reported as counterexamples have been checked by hand (the two new six-node graphs also by independent adversarial verification workflows, the 124 seven-node graphs also by four independently written exact verifiers). The 9,133 eight-node limit counterexamples have been verified by two independent machine campaigns and by *no* human hand check, and family members on $n \geq 8$ nodes are certified instances of proved theorems that have likewise *not* been hand-verified (Section 8); we label such results "machine-verified".

Reduction and exact machinery. At every legal parameter, the conjecture on n nodes is equivalent to the absence of a non-complete digraph on $k \leq n$ nodes whose full support is a positive, linearly stable fixed point (Theorem 2.6). The full-support problem has a rank-one structure: positivity depends on one parameter only, and the $\varepsilon \to 0$ limit is an exact spectral condition on the adjacency matrix compressed to $\mathbf{1}^{\perp}$ (Lemmas 2.2 and 2.4). Stability at some legal point is equivalent to stability on the one-dimensional legality curve (Lemma 3.4). Together with five necessary conditions valid at every parameter (domination, positivity, an edge bound, Stodola's condition and a twin obstruction, Section 3.4) these reduce the problem to a manageable set of exact one- and two-parameter certifications.

Complete maps for $n \leq 6$. No digraph on at most five nodes is a counterexample at any legal parameter. Among the 1,540,944 digraphs on six nodes exactly three non-complete graphs have a stable positive full support at some legal parameter: Geneson's graph, a partner obtained from it by a bidirected in-twin switch, and one further 18-edge graph (Theorem 4.2). The critical value is $q_6^* \approx 1.1664$, the root in $(561/481, 7/6)$ of $797q^6 - 1686q^5 + 2159q^4 - 3004q^3 + 2499q^2 - 2158q + 1521$ (Theorem 4.4).

The complete map at $n = 7$. Every one of the 882,033,440 digraphs on seven nodes is either excluded by a proved necessary condition or carries an exact two-parameter certificate. Exactly 124 non-complete graphs have a stable positive full support at some legal parameter (Theorem 5.1). Of these, 123 are visible in the $\varepsilon \to 0$ limit and one is not: its compressed adjacency has spectral abscissa exactly $-1/2$ and it is stable only in a window of moderate ε. The natural monotonicity statement "stable at finite ε implies stable in the limit" is false (Proposition 5.3), and an exact first-order coefficient R^* decides when a stable region bulges past its limit boundary (Theorem 5.4). The threshold is $q_7^* = \rho$, the plastic number (real root of $q^3 = q+1$), and for $n \leq 7$ the conjecture holds at every legal parameter with $q \geq \rho$ and fails for every $q \in (1, \rho)$ (Theorem 5.6).

The $\varepsilon \to 0$ map at $n = 8$. The edge bound restricts the search to the 801,558,826,511 digraphs on eight nodes with at least 29 edges, enumerated once each (the complete graph and three other high-symmetry graphs by a decomposition lemma, with exact ledgers). A C prefilter removes dominated graphs and graphs whose compressed adjacency has $\alpha \geq -1/2$, and the exact one-parameter engine classifies the 347,639 strict survivors: exactly 9,133 non-complete graphs are limit-stable with a positive fixed point (Theorem 6.1). The list has 6,260 classes of identical full-support data, 61 closed-twin lifts of seven-node counterexamples and 9,072 graphs that are not, 6,720 graphs containing no smaller counterexample, and the edge bound is attained by two 29-edge graphs. The limit threshold $q_8^*(\infty) \approx 1.7107$ is a root of an explicit degree-15 polynomial, exceeds the plastic number and is not attained (Theorem 6.2). All 9,133 are strongly connected, contain an induced oriented 3-cycle and have no bidirected clique on $n-2$ nodes; for limit counterexamples the last property is verified exhaustively up to $n = 9$ and stated as a conjecture (Conjecture 6.3). This is a limit map only: finite-ε-only counterexamples at $n = 8$, which exist, are outside it.

Infinite families and unbounded thresholds. Closed-twin blow-ups preserve full-support fixed-point data in closed form (Lemmas 7.1 and 7.2). This yields: counterexamples on every $n \geq 6$ (Theorem 7.3); finite-ε-only counterexamples on every $n \geq 7$ (Theorem 7.5); families whose thresholds approach 1.84, 3 and 3.96 (Theorem 7.6); and, via an in-tree construction realising prescribed limit polynomials, a proof that $q_n^* \to \infty$ (Theorem 7.7). Finally, at the single legal point $(\varepsilon, \delta) = (1/21, 11/210)$ there are stable non-clique full supports on n nodes for every $n \geq 6$, and on no $n \leq 5$ (Theorem 7.8).

Verification. Every count and critical value is backed by a re-runnable certificate file (Appendix A); the pipelines and theorems were independently audited; all $n \leq 7$ counterexamples were hand-verified, and the 124 seven-node ones were re-decided by four independently written exact verifiers; the 9,133 eight-node limit counterexamples were re-decided by four further exact verifiers written from the definitions and then by a second campaign of six lenses (completeness from the seven-node side, stability for all large t, the never side, the enumeration chain), with no human hand check; and 87 supporting theorems are machine-checked (Section 8).

1.4 Related work

The conjecture and the surrounding graph rules are from Curto and Morrison [3]. Graphical domination, which we use as a prune in its verbatim form, is due to Curto [2]. Geneson [4] gave the first counterexample and the bound (2); he does not pose or treat the gap question. His preprint has no later version at the time of writing (checked 28 October 2026), and we know of no published or posted work that classifies $n = 7$ or $n = 8$, exhibits finite-ε-only counterexamples or improves the bound: a search of arXiv on that date (topic searches for threshold-linear networks, CTLNs and target-free cliques, and author searches) found no CTLN item dated after 23 July 2026 apart from a revision of a preprint on fixed-point composition under gluings, which does not concern target-free cliques,

and the record and repositories of [1] were unchanged.[1] An unrefereed preprint, "Six Is Minimal" [1], obtained independently and by different exact methods that no digraph on at most five nodes has a stable non-clique support, used Geneson's graph as the six-node witness, computed the full-support region of that graph, and gave a family on $5r + 6$ nodes. Their conventions agree with ours up to a transpose of the non-edge matrix; their exclusion for $n \le 5$ agrees with our $n = 5$ map; their region for Geneson's graph gives $1 + u^* = 1.16635502785\ldots$, the same sextic root as our q_6^*; and their family is a clique-class blow-up of the same seed with $q \to 1$, an instance of Lemma 7.1. Their work does not classify $n = 6$ or treat $n \ge 7$, finite-ε-only examples, the reduction theorem or growing thresholds; it does give an explicit rational parameter schedule for its family and global nondegeneracy certificates, which prompted our nondegeneracy check. We regard the two works as independent and consistent. Our comparison is based on reading their repository; we did not run their code.

2 Preliminaries

2.1 Conventions

A digraph G on $[n]$ is encoded by its adjacency matrix A, with

$$A_{ij} = 1 \iff \text{there is an edge } i \to j \qquad (\text{row } i \text{ lists the nodes that } i \text{ sends to}).$$

The CTLN weight matrix is

$$W_{ij} = \begin{cases} 0, & i = j, \\ -1 + \varepsilon, & i \ne j,\ A_{ji} = 1 \quad (j \to i:\ \text{weak inhibition}), \\ -1 - \delta, & i \ne j,\ A_{ji} = 0 \quad (j \not\to i:\ \text{strong inhibition}). \end{cases} \tag{3}$$

Note the transpose: W_{ij} is the influence of j on i and therefore reads A_{ji}. For a support σ of size m with induced adjacency A_σ, with $J = \mathbf{1}\mathbf{1}^T$ the all-ones matrix,

$$I - W_\sigma \;=\; -\delta I + (1 + \delta)J - (\varepsilon + \delta)A_\sigma^T. \tag{4}$$

Every entry is a polynomial in (ε, δ) with integer coefficients. Identity (4), including the transpose, is machine-checked (`closed_form`$^{\llcorner}$).

Definition 2.1. A nonempty $\sigma \subseteq [n]$ *supports a stable fixed point* (SFP) if

(1) $(I - W_\sigma)x = \theta\mathbf{1}$ has a solution with $x > 0$ entrywise;

(2) $\theta + \sum_{j \in \sigma} W_{kj} x_j < 0$ for every $k \notin \sigma$;

(3) every eigenvalue of $-I + W_\sigma$ has negative real part.

If (1)–(3) hold, the zero-extension of x is an equilibrium of (1) near which the vector field is affine with Jacobian spectrum $\mathrm{spec}(-I + W_\sigma) \cup \{-1\}$, so it is linearly, hence locally asymptotically, stable. "Stable" always means linearly stable in this strict sense: points where $I - W_\sigma$ is singular or a Hurwitz determinant vanishes are never counted as stable. When $\sigma = [n]$, condition (2) is vacuous and "target-free clique" reduces to "complete bidirected".

Condition (3) is decided by the Hurwitz determinants of the exact characteristic polynomial: for a real $f(z) = c_0 z^m + \cdots + c_m$ with $c_0 > 0$, $\Delta_k(f) = \det[c_{2i-j}]_{i,j=1}^k$ (with $c_l = 0$ outside $0 \le l \le m$), and f has all roots in $\mathrm{Re}\, z < 0$ iff $\Delta_1, \ldots, \Delta_m > 0$ (Routh–Hurwitz; see Gantmacher, *The Theory of Matrices*, Vol. II, Ch. XV).

[1] `notes/literature_position_2026-10-28.md`.

2.2 Reparametrisation and legality

Put

$$t = \frac{1-\varepsilon}{\varepsilon}, \qquad \beta = 1 + \frac{\delta}{\varepsilon}, \qquad \gamma = 1 - \frac{1}{\beta}, \qquad q = \frac{\delta(1-\varepsilon)}{\varepsilon} = \frac{(\beta-1)t}{t+1}.$$

The map $(\varepsilon, \delta) \mapsto (\beta, t)$ is a homeomorphism of $(0,1) \times (0,\infty)$ onto $(1,\infty) \times (0,\infty)$ with inverse $\varepsilon = 1/(1+t)$, $\delta = (\beta-1)/(1+t)$, and

$$\text{legal} \iff q > 1 \iff t(\beta-2) > 1 \tag{5}$$

(`legal_iff`^{L}). In particular every legal point has $\beta > 2$ and $\gamma \in (1/2, 1)$, and along a ray $\delta = (\beta-1)\varepsilon$, $\varepsilon \to 0$, one has $t \to \infty$ and $q \to \beta - 1$. The drive $\theta > 0$ only rescales fixed points: each of (1)–(3), and every boundary event, has the same truth value at every $\theta > 0$ (Lemma G of the reduction note[2]).

Let $N := J - I - A_\sigma^T$ be the *receiver-first non-edge matrix*: $N_{ab} = 1$ iff $a \neq b$ and $\sigma_b \not\to \sigma_a$. Put $C := I + \beta N$ and $K := C + tJ$.

2.3 Rank-one structure

Lemma 2.2 (Rank-one structure; proved)**.** *Let $\varepsilon > 0$. Then*

(i) $I - W_\sigma = \varepsilon K$ *(`IminusW_eq`^{L});*

(ii) $a := \operatorname{adj}(K)\mathbf{1} = \operatorname{adj}(C)\mathbf{1}$ *for every value of t; each $a_i \in \mathbb{Z}[\beta]$ has degree $\leq m-1$;*

(iii) $\det K = d + ts$ *with $d = \det C$ and $s = \mathbf{1}^T a$;*

(iv) $\det(\mu I + K) = \phi(\mu) + t\,\omega(\mu)$ *with $\phi(\mu) = \det(\mu I + C)$ and $\omega(\mu) = \mathbf{1}^T \operatorname{adj}(\mu I + C)\mathbf{1}$, where ω has degree exactly $m-1$ in μ and leading coefficient m;*

(v) if $D := \det K \neq 0$, the solution of (1) is $x = (\theta/\varepsilon)\, a/D$, and for $k \notin \sigma$ the drive in (2) equals $\theta\, o_k/D$ with $o_k = d - \beta \sum_b N_{kb} a_b$, where $N_{kb} = 1$ iff $\sigma_b \not\to k$.

Sketch. (i) is an entrywise check using $\varepsilon\beta = \varepsilon + \delta$ and $\varepsilon t = 1 - \varepsilon$. (ii)–(iv) follow from the adjugate form of the matrix determinant lemma, $\det(M + \tau \mathbf{1}\mathbf{1}^T) = \det M + \tau\, \mathbf{1}^T \operatorname{adj}(M)\mathbf{1}$, valid over any commutative ring, and Cramer's rule in column form: in K with column i replaced by $\mathbf{1}$, subtracting t times that column from the others gives C with column i replaced by $\mathbf{1}$. (v) is Cramer's rule together with $W_{kb} = -\varepsilon(t + \beta N_{kb})$. No step divides by $\det C$. Full proofs: `notes/lemmas.md`, §1. □

Lemma 2.3 (Positivity is t-free; proved, ^{L})**.** *If K is positive stable (equivalently, (3) holds) then $\det K > 0$. Hence, at every point with $\varepsilon > 0$,*

$$\sigma \text{ is an SFP} \iff a_i(\beta) > 0 \ \forall i, \quad o_k(\beta) < 0 \ \forall k \notin \sigma, \quad \det(\mu I + K) \text{ Hurwitz.}$$

Sketch. $\det K$ is the product of the eigenvalues; real ones are positive and complex ones pair to $|\lambda|^2 > 0$. The Lean proof (`det_pos_of_posStable`, `ctln_pos_iff_adjugate_pos`) argues instead that $x \mapsto \det(xI + K)$ is monic with no root in $[0,\infty)$. □

The Lean development covers $\det K > 0$ and the positivity clause ($x > 0 \iff a > 0$ under stability); the off-support clause $o_k < 0$ is not formalised. Thus in the (β, t) quarter-plane the SFP set is a product of a set of β-intervals (positivity and the off-support condition) with $(0,\infty)$, intersected with the stability set and the legal region; only stability depends on t.

[2] `notes/lemma_F_reduction.md`, §1.

2.4 The $\varepsilon \to 0$ limit

Let $U \in \mathbb{R}^{m\times(m-1)}$ have orthonormal columns spanning $\mathbf{1}^\perp$ and write $X|_\perp := U^T X U$. Let $\alpha := \max \mathrm{Re}\,\mathrm{spec}(A_\sigma|_\perp)$ be the spectral abscissa of the compressed adjacency; it depends on the graph only. Exactly, $\mathrm{spec}\, A|_\perp$ can be computed over $\mathbb{Q}$ from any rational basis of $\mathbf{1}^\perp$.

Lemma 2.4 (Limit criterion; proved). *Let $\beta > 1$ and $\gamma = 1 - 1/\beta$.*

(a) $\omega(\mu;\beta) = m\det\big((\mu+1)I_{m-1} + \beta\, N|_\perp\big)$, and as $t \to \infty$ one eigenvalue of $K(\beta,t)$ is $mt + O(1)$ while the other $m-1$ converge to $\mathrm{spec}\big((1-\beta)I - \beta A|_\perp^T\big)$.

(b) If $\alpha < -\gamma$ there is T with $K(\beta,t)$ positive stable for all $t > T$; if $\alpha > -\gamma$, then $K(\beta,t)$ is unstable for all large t; $\alpha = -\gamma$ is undecided by the limit. The limit-stable set of legal β is the interval $(2,\beta_c)$, $\beta_c = 1/(1+\alpha)$, nonempty iff $\alpha < -1/2$; always $\alpha \geq -1$, with equality iff $G[\sigma]$ is complete bidirected. The threshold T is uniform on compact subsets of $(2,\beta_c)$.

(c) $a(\beta)$ is one-signed $\iff$ $A^T + \gamma I$ is nonsingular and $w := (A^T+\gamma I)^{-1}\mathbf{1} > 0$; then a is a nonzero multiple of w, and condition (1) holds at (β,t) for some, equivalently every, $t > 0$ iff $w > 0$. For $k \notin \sigma$, $o_k s < 0$ iff $\sum_{b:\sigma_b \to k} w_b < 1$.

(d) If $\alpha < -\gamma$ at a legal β, then $G[\sigma]$ has more than $m(m-1)/2$ edges.

Sketch. (a) Conjugating by the orthogonal matrix $[\mathbf{1}/\sqrt{m}\,|\,U]$ turns tJ into $tm\, e_1 e_1^T$, and the determinant is affine in the $(1,1)$ entry with slope the complementary minor; root continuity (Rouché, or an elementary continuity lemma) gives the eigenvalue statement. (b) follows from (a) and $J|_\perp = 0$, since $N|_\perp = -I - A|_\perp^T$. (c) uses $Ca = d\mathbf{1}$ and $C = -\beta(A^T+\gamma I) + \beta J$, so $(A^T+\gamma I)a = \frac{\beta s - d}{\beta}\mathbf{1}$, together with $\det(A^T+\gamma I) = (-1/\beta)^m(d - \beta s)$. (d) is a trace computation: $\mathrm{tr}\, A|_\perp = -|E|/m$ and the maximum real part dominates the mean. Full proofs: `notes/lemmas.md`, §4, and `notes/lemma_I_spectral.md`, §§2–6. □

Combining (b) and (c): if $w(\gamma) > 0$ and $\alpha < -\gamma$ at some rational $\beta > 2$, then the full support is an SFP at the legal points $(\varepsilon,\delta) = (1/(1+t), (\beta-1)/(1+t))$ for all $t > T(\beta)$. We call such a graph *limit-stable*. Lemma 2.4 is the shared dependency of every "for all k" statement in Section 7; it is not formalised in Lean. Parts (b) and (c) are Lemmas I.3 and I.4 of the project notes; they were reviewed internally when the $n = 6$ map was audited and were not re-derived by the internal reviewers of the family theorems.

2.5 Reduction to full supports

Lemma 2.5 (Cliques; proved). *Let $0 < \varepsilon < 1$, $\delta > 0$, $\theta > 0$ and let σ be a clique of size m. Then $x = c\mathbf{1}$ with $c = \theta/(\varepsilon + m(1-\varepsilon))$ solves (1), (3) holds, and the drive at $k \notin \sigma$ is $\theta[\varepsilon - r(\varepsilon+\delta)]/(\varepsilon + m(1-\varepsilon))$ where r is the number of $j \in \sigma$ with $j \not\to k$. Hence a clique is an SFP iff it is target-free.*

For a legal point p and $k \geq 2$ let $\mathrm{FS}(k,p)$ be the statement "every k-node digraph whose full support satisfies (1) and (3) at p is complete bidirected".

Theorem 2.6 (Reduction to full supports; proved). *Fix a legal $p = (\varepsilon,\delta)$ and $n \geq 1$. The following are equivalent:*

(i) the target-free clique conjecture holds for every n-node CTLN at p;

(ii) for every n-node G, every SFP support of G at p is a clique;

(iii) $\mathrm{FS}(k,p)$ holds for every $k \in \{2,\dots,n\}$.

Sketch. (i)⇔(ii) by Lemma 2.5. (iii)⇒(ii): conditions (1) and (3) for σ in G involve only W_σ, which is the weight matrix of the induced subgraph $G[\sigma]$; so a non-clique SFP σ would give a non-complete $|\sigma|$-node graph satisfying (1) and (3) on its full support. (ii)⇒(iii) by padding: given a non-complete H on $k \le n$ nodes with (1) and (3) on its full support, add $n-k$ nodes receiving no edges from H. At such a node the drive is $\theta - \sum_j c_{kj} x_j$ with all $c_{kj} = 1+\delta$; subtracting row i of the fixed-point equation shows it is at most $-\delta x_i < 0$, so (2) holds and $V(H)$ is a non-clique SFP of the padded graph. Condition (2) enters only to exclude targets of cliques. Full proof: `notes/lemma_F_reduction.md`, §6 (there called Theorem F). □

Corollary 2.7. *Let* $\mathrm{Bad}(n)$ *be the set of legal points at which the conjecture fails for some* n*-node graph, and* S_H *the set of legal points at which the full support of* H *is an SFP. Then* $\mathrm{Bad}(n) = \bigcup_{k\le n} \bigcup_H S_H$ *over non-complete* k*-node* H*, and* $\mathrm{Bad}(n) \subseteq \mathrm{Bad}(n+1)$.

Lemma 2.8 (Openness; proved)**.** *Each* S_H *is open in the* (ε, δ) *plane. Consequently every nonempty* S_H *contains rational points, and at every point of* S_H *all Hurwitz determinants and coefficients of the characteristic polynomial are strictly positive: a counterexample point is never an "undecided" point.*

We write q_n^* for the supremum of q over legal points at which some non-complete n-node digraph has a stable positive full-support fixed point. By Corollary 2.7, the conjecture for all graphs on at most n nodes holds at every legal point with $q \ge \max_{k\le n} q_k^*$ and fails at points with q arbitrarily close to that maximum. For $k \le 4$ there is no counterexample at any legal parameter: $k = 2$ by a direct eigenvalue computation (for a one-way edge $\det K = 1 - t(\beta - 2)$, negative exactly at legal points; a non-adjacent pair is excluded by Lemma 3.8), $k = 3$ by Theorem 4.1 of [4], $k = 4$ is stated as known in [4] (for nondegenerate CTLNs, citing work of Curto, Geneson and Morrison whose exact hypotheses we have not checked), and all three cases are re-certified by our two-parameter engine on all digraphs with at most four nodes; our claims for $k \le 4$ rest on that certification.[3]

3 The exact machinery

3.1 Boundary sets and three-valued decisions

At a single sampled parameter a vanishing Hurwitz determinant says nothing about the neighbourhood, so every pointwise procedure in this work returns one of three values: *stable*, *not stable*, or *undecided*. An undecided sample is never counted as unstable and never as a counterexample. Regions are decided instead on open cells of a decomposition whose walls are the zero sets of a proved boundary set.

Lemma 3.1 (Imaginary-axis roots; proved)**.** *If* f *is real of degree* $m \ge 2$ *and* $f(iw) = 0$ *for some real* $w \ne 0$*, then* $\Delta_{m-1}(f) = 0$.

Sketch. With $P(x) = x^m f(1/x)$, the factor $1 + w^2x^2$ divides $P = (1+w^2x^2)R$; for $V(x) = xR(-x) \ne 0$ the product PV is odd, and its even coefficients are exactly the entries of $H_{m-1}v$ where H_{m-1} is the matrix whose determinant is Δ_{m-1}. This is the special case of Orlando's formula that is needed; it is proved self-contained in `notes/lemmas.md`, §2. □

Lemma 3.2 (Boundary set at fixed t; proved)**.** *Fix* $t > 0$ *and let* $p_t(\mu;\beta) = \det(\mu I + K)$*. Let* P_t *be the family* $a_1, \dots, a_m$*,* $D_t = d + ts$*,* o_k $(k \notin \sigma)$*,* $\Delta_{m-1}(p_t)$*, all in* $\mathbb{Q}[\beta]$*. If some member is identically zero,* σ *is an SFP at no legal* β*. Otherwise the SFP verdict is constant on each component of the legal* β*-range minus the real roots of* P_t*, and false at those roots. The same holds with* β *fixed and* t *varying, and in the limit* $t = \infty$ *with* ω*,* s*,* $\Delta_{m-2}(\omega)$ *in place of* p_t*,* D_t*,* $\Delta_{m-1}(p_t)$.

[3] `certificates/region2d_n2to4_all.json`; `notes/lemma_F_reduction.md`, §§5, 8.

Sketch. On a component, every member of P_t has constant sign; by Lemma 3.1 no root of p_t meets the imaginary axis; by continuity of roots the number of roots in the right half-plane is locally constant. `notes/lemmas.md`, §2. □

The two-parameter version (Lemma B′ and Corollaries B″, B‴ of the project notes[4]) shows that the verdict is invariant on connected components of the complement of the curves $\{a_i = 0\}$, $\{D = 0\}$, $\{\Delta_{m-1} = 0\}$ and the legality boundary $L(\beta, t) = t(\beta - 2) - 1 = 0$ in the open legal region, and that over each open β-interval free of the roots of a projection set (leading coefficients, discriminants and pairwise resultants of these polynomials in t) the sectors of a cylindrical algebraic decomposition are sign-invariant cells. Each full-dimensional cell is decided at one exact rational sample; lower-dimensional cells are boundary points and are never SFP.

3.2 The engines

One-parameter engine. At fixed t (or $t = \infty$), the polynomials of Lemma 3.2 are obtained by exact integer evaluation and interpolation with provably sufficient degree bounds, their real roots are isolated by continued fractions into irreducible minimal polynomials with rational isolating intervals, and each cell is decided at an exact sample by the Hurwitz routine. Every isolation is replayed by Sturm counts and every sample is cross-checked against an independent pointwise verifier. The engine agrees with an independent symbolic path on all 7,194 (graph, support, ε) cases at $n = 4$ and on 900 at $n = 5$.[5]

Two-parameter engine. A cylindrical algebraic decomposition of the legal (β, t) plane on the proved boundary set, restricted to the positivity cells, with a Descartes-bisection replay of every root count that is independent of the isolator. It reproduces the theorem for $k \le 4$ on all digraphs with at most four nodes and agrees with an earlier implementation on all 706 five-node survivors of a weaker filter.[6]

Policy. A cell whose Hurwitz test is undecided at three samples is *escalated*, not decided; there were 0 escalations at $n = 5, 6, 7, 8$. By Lemma 2.8 undecided points cannot hide counterexamples.

3.3 Ray reduction to the legality curve

Lemma 3.3 (Ray identity; proved, ᴸ). *$K(\lambda\beta, \lambda t) = (1 - \lambda)I + \lambda K(\beta, t)$. Hence if $K(\beta, t)$ is positive stable, so is $K(\lambda\beta, \lambda t)$ for every $\lambda \in (0, 1]$ (`affine_identity`, `posStable_ray`ᴸ).*

Lemma 3.4 (Ray reduction; proved, forward direction ᴸ). *The full support is stable at some legal (ε, δ) if and only if $K(2 + 1/t,\, t)$ is positive stable for some $t > 0$. Moreover, if the full support is an SFP at a legal (β_0, t_0), the corresponding curve point has $\beta' < \beta_0$.*

Proof. ($\Leftarrow$) Positive stability is open, so $K(2 + 1/t + \eta, t)$ is stable for small $\eta > 0$, and that point is legal. ($\Rightarrow$) Let (β, t) be legal and stable and $f(\lambda) = \lambda t(\lambda\beta - 2) - 1$. Then $f(2/\beta) = -1 < 0 < f(1)$, so f has a root $\lambda^* \in (2/\beta, 1)$, explicitly $\lambda^* = (t + \sqrt{t^2 + \beta t})/(\beta t)$; the point $(\lambda^*\beta, \lambda^* t)$ lies on the curve $t(\beta - 2) = 1$ and is stable by Lemma 3.3. The forward direction is `lemmaR_forward_ctln`ᴸ; the converse needs continuity of eigenvalues and is not formalised. □

Since positivity is t-free (Lemma 2.3), a graph that is never stable on the curve has no SFP anywhere in the legal plane. Along the curve, $tK(2+1/t, t) = L(t) := tI + (2t+1)N + t^2 J$ is an integer polynomial matrix; the coefficient of μ^{m-k} in $\det(\mu I + L(t))$ has t-degree at most $k + 1$, and by Lemma 3.2 with

[4] `notes/lemma_Bprime_2d.md`.

[5] `certificates/validate_fastregion_n4.json`, `certificates/validate_fastregion_n5.json`.

[6] `certificates/region2d_n2to4_all.json`, `certificates/region2d_v2_validation.json`.

t as the parameter the verdict changes only at positive roots of $\det L(t)$ and of $\Delta_{m-1}(t)$. One exact sample per root-free interval decides the curve (the *curve test*). A fast path discards a graph when no $t > 0$ makes every coefficient positive (Stodola's condition, below). The curve test takes about 35–45 ms per seven-node graph, and on all 9,608 five-node digraphs returns 95 stable and 9,513 never, each of the 95 with an exact legal witness.[7]

3.4 Necessary conditions valid at every parameter

Every prune below is a theorem valid at *every* legal parameter, and all five are machine-checked. The twin obstruction is not a separate stage of the $n = 7$ chain.

Lemma 3.5 (Domination; proved, ᴸ). *Node k* graphically dominates *$j \neq k$ if $j \to k$, $k \not\to j$, and every $i \notin \{j, k\}$ with $i \to j$ also has $i \to k$ [2, Def. 1.1]. If k dominates j, then every fixed point of (1) with uniform drive has $x_j = 0$, for every $\varepsilon > 0$ and $\delta \geq 0$; in particular the full support has no positive fixed point. The same holds for domination inside an induced subgraph $G[\sigma]$ (`lemmaD`, `lemmaD_full_support`, `lemmaD_induced`ᴸ).*

Sketch. For $x \geq 0$ one has $y_j(x) - y_k(x) \leq (1-\varepsilon)x_j - (1+\delta)x_k$, where y_l is the input to node l. At a fixed point with $x_j > 0$ this forces $y_k > 0$ and then $0 \geq \varepsilon x_j + \delta x_k > 0$. Neither $\varepsilon < 1$ nor legality is used, but the uniform drive is: with $\theta_j > \theta_k$ the conclusion fails. `notes/lemmas.md`, §5. □

Lemma 3.6 (Edge bound; proved, ᴸ). *Let B, N_n, O be the numbers of bidirected, non-adjacent and one-way unordered pairs of an m-node graph. The sum of the 2×2 principal minors of K is*

$$e_2(K) = B(1+2t) + N_n(1-\beta)(1+\beta+2t) + O\big(1 - t(\beta-2)\big).$$

If K is positive stable at a legal point then $e_2(K) > 0$, hence $B > N_n$ and the graph has at least $m(m-1)/2 + 1$ edges (`e2_closed_form`, `edge_bound_ctln`ᴸ).

At a legal point the non-adjacent and one-way terms are negative, which gives $B > N_n$. In Lean the spectral step (positive stability forces $e_2 > 0$) is proved from Mathlib's description of characteristic-polynomial coefficients as sums of principal minors and the Gauss–Lucas theorem. For $n = 7$ the bound reads "at least 22 edges".

Lemma 3.7 (Stodola; proved, ᴸ). *A monic real polynomial with all roots in $\operatorname{Re} z < 0$ has all coefficients positive. Hence if K is positive stable, every coefficient of $\det(\mu I + K)$ is positive (`stodola`, `coeff_pos_of_posStable`, `ctln_coeff_pos`ᴸ).*

Lemma 3.8 (Twin obstruction; proved, ᴸ). *If $u, v \in \sigma$ are non-adjacent and have the same in-neighbours in $\sigma \setminus \{u, v\}$, then $e_u - e_v$ is a left eigenvector of W_σ with eigenvalue $1 + \delta$, so $-I + W_\sigma$ has the eigenvalue $\delta > 0$ and σ is never stable, for every ε and every $\delta > 0$ (`twin_not_hurwitz`, `twin_not_hurwitz_induced`ᴸ). For bidirected in-twins the eigenvalue is $-\varepsilon$.*

Together with Lemma 2.3 (a positive stable full support needs a legal β with all $a_i(\beta) > 0$) and the density bound Lemma 2.4(d) for the limit, these are the only prunes used. An early placeholder for domination was replaced by the verbatim definition of [2] before any sweep used it.

[7] `n7/validate_curve_n5.json`; `notes/lemma_M_refuted.md`, §3.

4 Complete maps for $n \leq 6$

4.1 Validation gates

Two published results were reproduced before any new computation was trusted.

Geneson's counterexample. The paper [4] gives its graph as a receiver-first non-edge matrix; our adjacency matrix is its complement and transpose, and the weight convention is the same. The printed integer certificate $Cv = 179298949 \cdot \mathbf{1}$ reproduces exactly, our fixed point is proportional to the paper's v, and the full support is a stable non-clique fixed point at $\varepsilon = 1/100$, $\delta = 29/2500$ and at $\varepsilon = 1/1000$, $\delta = 29\varepsilon/25$, while it is a positive but unstable fixed point at $\varepsilon = 1/10, 1/20, 1/50$ on the same ray.[8]

The published $n = 5$ classification (the package accompanying Curto's $n = 5$ data, which stores the transpose of our adjacency matrix). At its default parameters $\varepsilon = 1/4$, $\delta = 1/2$, every recorded fixed-point support of every one of the 9,608 five-node digraphs matches our exact verifier: 24,442 supports, 0 disagreements, 0 boundary cases (the package records no stability data; our verifier finds 14,488 of the supports stable). Under the wrong (untransposed) convention a sampled version of this check fails on 27 of 30 graphs.[9]

Our enumerations reproduce the number of digraphs up to isomorphism (OEIS A000273): 9,608 for $n = 5$, 1,540,944 for $n = 6$ and 882,033,440 for $n = 7$.

4.2 $n \leq 5$

Theorem 4.1 ($n \leq 5$; certified)**.** *No non-complete digraph on at most five nodes has a stable positive full-support fixed point at any legal parameter. Consequently, by Theorem 2.6, the target-free clique conjecture holds for every CTLN on at most five nodes at every legal parameter.*

Of the 9,608 five-node digraphs, 390 admit a legal β with all $a_i(\beta) > 0$; the two-parameter decomposition certifies 389 of them as never stable and only K_5 as stable, with 0 escalations. (An earlier, weaker common-sign filter left a superset of 706, which was also certified in full: 705 never, K_5 stable.)[10] This agrees with the independent result of [1].

4.3 $n = 6$

Theorem 4.2 ($n = 6$; certified, hand-verified)**.** *Among the* 1,540,944 *digraphs on six nodes, exactly three non-complete graphs have a stable positive full-support fixed point at some legal parameter (Table 1). Every other six-node graph is certified never stable and positive at every legal* (ε, δ)*.*

graph	adjacency rows	edges	limit-stable q	relation
G_1 (index 1,000,255)	`011000 000011 000111 110011 110101 101110`	19	$(1, q_6^*)$	Geneson
G_2 (index 1,424,629)	`011000 000101 000111 100011 110101 111110`	19	$(1, q_6^*)$	switch of G_1
H_3 (index 1,387,528)	`011000 000011 000111 110010 100101 111110`	18	$(1, q_c')$	new

Table 1: The three six-node counterexamples, in canonical labelling (row i lists the j with $i \to j$); G_2 is a bidirected in-twin switch of G_1. $q_c' \approx 1.0830$ is the root in $(1, 13/12)$ of $q^5 + 2q^4 + q^3 - 2q^2 - 2q - 1$.

Figure 1 shows Geneson's graph and H_3; G_1 and G_2 differ from Geneson's graph only by relabelling and by a twin switch (below).

[8] `certificates/stage0_geneson_2607.21396.json`; a permanent test in the project's test suite.

[9] `certificates/stage1_n5_diff.json`.

[10] `certificates/region2d_n5_shard0of1_tally.json`, `certificates/n5_summary.json`.

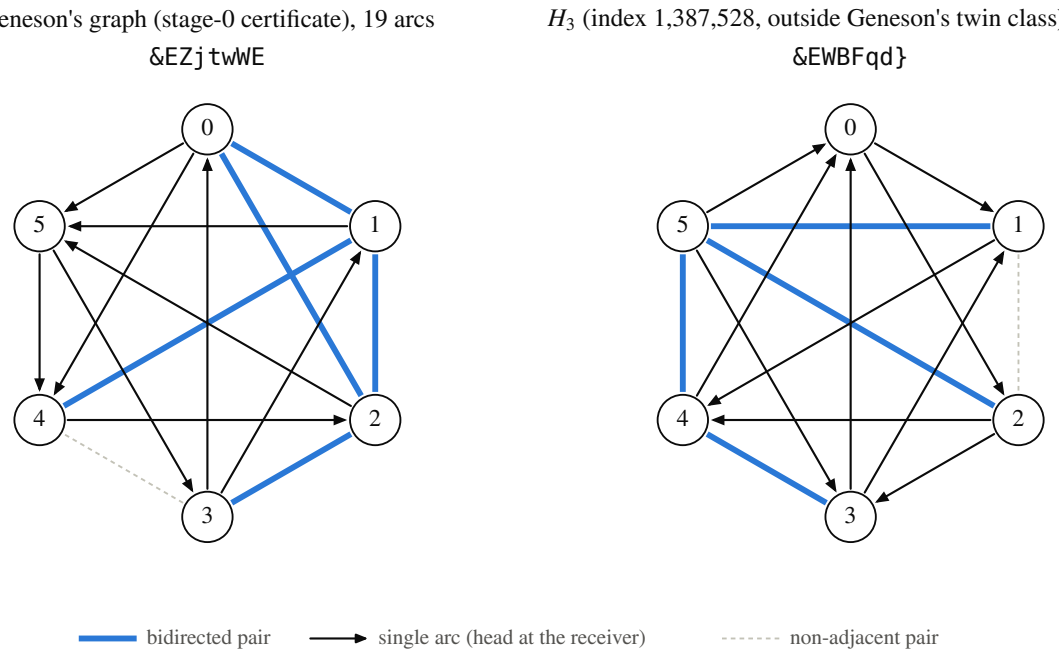


Figure 1: Two of the six-node counterexamples. Left: Geneson's graph in the labelling of the stage-0 certificate (`certificates/stage0_geneson_2607.21396.json`; 19 arcs; it is isomorphic to G_1 of Table 1). Right: H_3 (index 1,387,528, 18 arcs, adjacency from `certificates/n6_summary.json`), the counterexample outside Geneson's twin class. Vertices are placed on a circle in digraph6 order, clockwise from the top; thick segments are bidirected pairs, thin arrows are single arcs with the head at the receiver, light dashed segments are non-adjacent pairs. The digraph6 string above each panel encodes the adjacency matrix row by row (row i lists the j with $i \to j$). Drawn by `notes/paper/figures/make_figures.py`, which records every drawn arc in `notes/paper/figures/figures_data.json`.

The chain. In the limit sweep over all 1,540,944 graphs, 1,512,594 have no legal β with all $a_i > 0$ and therefore (Lemma 2.3) no stable positive full support at any parameter; 28,350 strict survivors remain, all among the 204,378 domination-free graphs; exactly four are limit-stable (K_6, G_1, G_2, H_3); 0 escalations. The two-parameter decomposition was run on a superset of 34,071 survivors (an earlier weaker filter): 34,067 never stable anywhere, 4 stable somewhere, 0 escalations, 0 errors, every cell cross-checked pointwise and every root count replayed. Exact β-slices at $\varepsilon = 1/1000$ and $1/100$ show K_6 and the three graphs; at $\varepsilon = 1/10$ only K_6 and Geneson's class; at $\varepsilon = 1/4$ and $1/2$ only K_6.[11]

The regions. For G_1 and G_2 the stable positive region is $\{2 < \beta < \beta_c,\ t > \max(1/(\beta-2), T(\beta))\}$ with $T(\beta) < 1/(\beta-2)$ on an initial interval $(2, \beta_1)$, β_1 a degree-14 algebraic number in $(17/8, 15/7)$: these graphs are stable at *every* legal t when δ/ε is close enough to 1, so counterexamples exist with q arbitrarily close to 1. For H_3 the two thresholds coincide: the full support is an SFP iff $2 < \beta < \beta_1'$, at every legal t, where $\beta_1' \approx 2.08302$ is the root in $(152/73, 25/12)$ of $\beta^5 - 3\beta^4 + 3\beta^3 - 3\beta^2 + 2\beta - 1$, a positivity boundary ($a_5 = 0$). A witness is $\varepsilon = 1/101$, $\delta = 27/2626$, $q = 1350/1313$.[12]

[11] `certificates/n6_summary.json`; `certificates/region2d_n6_shard{0,1,2}of3*`; `certificates/region_n6_eps*`.

[12] `certificates/COUNTEREXAMPLE_CANDIDATE_n6_idx1387528.json`, `certificates/COUNTEREXAMPLE_CANDIDATE_n6_idx1424629.json`.

Twins. G_2 is obtained from G_1 by moving one out-edge between a pair of bidirected in-twins.

Proposition 4.3 (Bidirected in-twin switching; proved)**.** *Let $u \leftrightarrow v$ be bidirected and have the same in-neighbours outside $\{u, v\}$, and let G' be obtained from G by replacing an edge $u \to w$ (with $v \not\to w$) by $v \to w$. Then for every support, every $\varepsilon, \delta > 0$ and every θ, G and G' have the same fixed points (same vectors) and the same Jacobian characteristic polynomials.*

More generally, two graphs have identical full-support data for all parameters iff after relabelling they are cospectral and have the same in-walk counts $(A^T)^k\mathbf{1}$ for all k; the 34,071 six-node survivors form 21,800 such classes. Bidirected in-twin switching preserves all supports, whereas other members of a class can differ in their proper supports.[13] So modulo twin switching there are exactly two six-node counterexample graphs.

4.4 The threshold at $n \le 6$

In the limit, $\beta_c(G_1) = -1/\operatorname{Re}\nu_1$, where ν_1 is a root of the quartic factor $6z^4 + 11z^3 + 16z^2 + 10z + 3$ of the characteristic polynomial of the compressed non-edge matrix (the other factor is z, from a pair of twins). Real parts of roots of a quartic are roots of a polynomial of degree at most $\binom{4}{2} = 6$; here β_c is a root of the irreducible sextic $797b^6 - 6468b^5 + 22544b^4 - 44440b^3 + 53280b^2 - 38016b + 13824$, and the substitution $b = q + 1$ gives the sextic in q below.

Theorem 4.4 (Threshold for $n \le 6$; certified)**.** *Let q_6^* be the root in* (561/481, 7/6) *of*

$$797q^6 - 1686q^5 + 2159q^4 - 3004q^3 + 2499q^2 - 2158q + 1521 \qquad (q_6^* \approx 1.1664).$$

For every graph on at most six nodes, the target-free clique conjecture holds at every legal parameter with $q \ge q_6^*$; for every $q \in (1, q_6^*)$ there is a legal parameter with that value of q at which it fails for a six-node graph. The supremum is approached by G_1 and G_2 as $t \to \infty$ and is not attained.

Sketch. By Theorem 4.1, Theorem 4.2 and Corollary 2.7, $\mathrm{Bad}(6) = S_{G_1} \cup S_{G_2} \cup S_{H_3}$. On every finite-$t$ cell $q = (\beta - 1)t/(t + 1) < \beta - 1$, and the exact suprema over the certified cells are computed by the threshold tool from the two-parameter certificates; the limit interval of G_1 starts at $q = 1$, so the union covers $(1, q_6^*)$.[14] □

Geneson's bound (2) at $n = 6$ reads $q \ge 4 - 3\varepsilon/2$; Theorem 4.4 shows the true value is $q_6^* \approx 1.1664$. The same sextic root, 1.16635502785…, was obtained independently in [1].

5 The complete map at $n = 7$

5.1 The accounting chain

Theorem 5.1 ($n = 7$; certified, hand-verified)**.** *Among the* 882,033,440 *digraphs on seven nodes, exactly* 124 *non-complete graphs have a stable positive full-support fixed point at some legal (ε, δ). Each carries an exact two-parameter region certificate. Of the* 124*, exactly* 123 *are limit-stable; the remaining graph,* `&FNFL~[udo?`*, is stable only at finite ε.*

Every digraph is either removed by a necessary condition valid at every legal parameter, or certified by the two-parameter engine (Table 2). Each stage was validated before use against the $n = 6$ map (the positivity stages reproduce the 28,350 six-node strict survivors exactly, with zero difference in both directions; the curve test applied to those survivors returns exactly the four stable six-node graphs).

[13] `notes/lemma_J_equivalence.md`; `certificates/n6_survivor_equivalence_classes.json`.

[14] `certificates/qstar/qstar_n6.json`, tool `n7/qstar.py`.

stage	condition (where proved)	removed	remaining
0	enumeration (OEIS A000273)		882,033,440
1	a dominated node (Lemma 3.5)	764,800,518	117,232,922
2	no $\beta > 2$ with all $a_i(\beta) > 0$, exact Descartes bounds (Lemma 2.3)	107,600,789	9,632,133
3	undecided lines of stage 2 resolved by exact root isolation	723,010	8,909,123
4	at most 21 edges (Lemma 3.6)	5,135,775	3,773,348
5	never stable on the legality curve (Lemma 3.4)	3,773,183	165
6	stable somewhere, never stable and positive together (CAD)	40	125

Table 2: Accounting chain at $n = 7$. The 125 survivors are K_7 and the 124 counterexamples; 0 escalations at every stage.

Of the 9,632,133 lines kept at stage 2, 8,845,537 were certified positive and 786,596 were undecided by the fast exact filter; stage 3 resolved the latter exactly. The single run performing stages 4 and 5 over the 8,909,123 positivity survivors took 16.4 CPU-hours (6.6 ms per line).[15] Independently, the limit classification was run first: a spectral prefilter over all 882,033,440 digraphs left 638,042 survivors (1,889 strict), whose exact limit classification gave the 123 limit-stable non-clique graphs and K_7.[16] The finite-ε map contains all of them, as Lemma 2.4 requires.

5.2 The 124 graphs

- Edge counts range from 23 to 31 of 42 possible (counts $7, 7, 18, 10, 6, 22, 34, 17, 3$ for $23, \ldots, 31$ edges); all satisfy the edge bound (≥ 22) and all are domination-free.
- They form 91 classes of identical full-support data (Section 4): 71 singletons, 16 pairs, two classes of 4, one of 5 and one of 8; 64 graphs carry a bidirected in-twin pair.
- Of the 123 limit-stable graphs, the limit critical values q_c (right ends of the limit-stable q-intervals) have 69 distinct minimal polynomials; the most frequent is $q^5+2q^4+q^3-2q^2-2q-1$ (17 graphs, the same quintic as for H_3), and the largest q_c is the plastic number, the real root of $q^3 - q - 1$ (6 graphs). For 6 of the 123 the positivity set starts strictly above $\beta = 2$.
- 21 of the 123 contain a six-node counterexample as an induced subgraph; none contains K_6.

Every graph comes with a legal exact witness point.[17]

Corollary 5.2 (certified)**.** *Every seven-node graph whose full support is an SFP at some legal parameter has $\alpha \leq -1/2$, with equality only for* `&FNFL~[udo?`*.*

5.3 A counterexample invisible in the limit

The graph E_0 = `&FNFL~[udo?` (28 edges) has adjacency rows `0011110 0011100 1101111 1110111 0011011 0100101 1100000`. Its compressed adjacency satisfies

$$7\chi_{A|_\perp}(\lambda) = (\lambda+1)(\lambda^2+\lambda+1)(7\lambda^3+14\lambda^2+14\lambda+5),$$

[15] `certificates/n7_finite_summary.json` (field `chain`, `accounting_ok = true`); `certificates/n7_curve_TOTAL.json`; `certificates/n7_region2d/`; runbook `n7/REPRODUCE_n7_finite.md`.

[16] `certificates/COUNTEREXAMPLE_CANDIDATES_n7_limit.jsonl`, `certificates/n7_limit_tally.json`. For 15 graphs the first stored pointwise witness in the limit file is an illegal point ($q \leq 1$); use `certificates/n7_candidates_legal_witnesses.json`.

[17] `certificates/COUNTEREXAMPLE_CANDIDATES_n7_finite.jsonl`; `certificates/n7_candidates_legal_witnesses.json`.

so $\alpha = -1/2$ exactly and, by Lemma 2.4(b), the graph is limit-stable at no legal β. Nevertheless its full support is a stable positive fixed point at $\varepsilon = 1/49$, $\delta = 33/1568$ ($q = 99/98$) and at $\varepsilon = 1/36$, $\delta = 17/588$. Its stable region is bounded in t at every β: on its single positivity cell $2 < \beta < \beta_1'$ ($\beta_1' \approx 2.08302$, the quintic root of Table 1, isolated here in $(27/13, 25/12)$), at the sample $\beta = 53/26$, the stable range is $26 < t < t_1$ with t_1 an algebraic number in $(53, 54)$, and the full support is unstable for larger t. The supremum of q over its region is an algebraic number of degree 25 in (685003/657781, 2170535/2084278), about 1.0414, approached at the corner where $\Delta_6 = 0$ meets the positivity boundary, not as $t \to \infty$.[18] Structurally, E_0 is H_3 with one node doubled into a closed-twin pair, and the factor $\lambda^2 + \lambda + 1$ is a *moving* eigenvalue pair (not an eigenvalue of A) that sits exactly on the boundary line; see Theorem 7.5.

5.4 The limit does not determine the finite map

Write $c = 1 + t/\beta > 1$. A one-line computation gives the identity

$$K = \beta\,(cJ - A^T - \gamma I), \tag{6}$$

so K is positive stable iff $\nu_A(c) := \min \mathrm{Re}\,\mathrm{spec}(cJ - A) > \gamma$, and legality is $\gamma > c - \sqrt{c^2 - c}$, which is the stability boundary of a single one-way edge. As $c \to \infty$, $\nu_A(c) \to -\alpha$. The natural hope is that ν_A is nondecreasing, i.e. that stability at a legal finite t implies stability for all larger t at the same β ("Lemma M").

Proposition 5.3 (Lemma M is false; certified)**.** *(a) For A with rows `011` `001` `100` and $\beta = 22/5$, K is positive stable at the legal point $t = 11/25$ and not at $t = 1$ or $t = 100$; here $\alpha = -2/3 > -17/22 = -\gamma$. (b) For the domination-free seven-node graph `&FK@dyUfXp?` (one of the 123) at $\beta = 10561/5000$, the full support is an SFP at $t = 10$ and positive but unstable at $t = 1000$, and the limit polynomial is not Hurwitz.*

Example (a) is not a CTLN counterexample (it fails positivity), but (b) shows the failure persists with positivity. In the latter, the right end of the stable β-interval at $t = 10, 30, 100, 1000, 10^4$ lies strictly above the limit value β_c.[19] Lemma 2.3 shows that every such failure under positivity is an imaginary-pair crossing: on the positivity set $s > 0$, so $D = d + ts$ increases in t and a real eigenvalue crosses 0 only upward.

5.5 The bulge criterion

Let $\chi(z) = \det(zI + A)$, $\psi(z) = \mathbf{1}^T \mathrm{adj}(zI + A)\mathbf{1}$ and $\hat{s} = \psi/\chi = \mathbf{1}^T(zI + A)^{-1}\mathbf{1}$. Then $\det(zI - (cJ - A)) = \chi(z) - c\psi(z)$; roots of $\gcd(\chi, \psi)$ are eigenvalues for every c (*fixed*), the others are *moving*.

Theorem 5.4 (First-order asymptotics; proved)**.** *Assume $\nu_\infty := -\alpha \in (1/2, 1)$ and that every root z_0 of ψ with $\mathrm{Re}\, z_0 = \nu_\infty$ is simple and not a root of χ. Let $R^* = \min \mathrm{Re}\big(1/\hat{s}\,'(z_0)\big)$ over these roots. Then $\nu_A(c) = \nu_\infty + R^*/c + O(c^{-2})$, and the boundary of the stable region near (β_c, ∞) is $\beta = \beta_c + \beta_c^3 R^*/t + O(t^{-2})$.*

(a) If $R^* > 0$, then for every β slightly above β_c the stable t-set is nonempty and bounded above; if moreover $a(\beta_c) > 0$, these points are SFPs.

(b) If $R^* < 0$, the stable t-set is upward closed for large t near β_c.

[18] `certificates/COUNTEREXAMPLE_CANDIDATE_n7_finite_FNFL.json`; `certificates/qstar/qstar_fnfl.json` (the exact bracket is recorded there and in `qstar_n7_finite.json`; the file's aggregate block is stale and reports only a weaker sample bound).

[19] `notes/lemma_M_refuted.md`, §2; scripts in `notes/prover_scripts_M/`.

Sketch. With $e = 1/c$ the moving eigenvalues solve $\psi_1(z) - e\chi_1(z) = 0$ ($\chi = h\chi_1$, $\psi = h\psi_1$, $h = \gcd$); the implicit function theorem gives an analytic branch $z(e) = z_0 + e\,\chi_1(z_0)/\psi_1'(z_0) + O(e^2)$, and $\chi_1(z_0)/\psi_1'(z_0) = 1/\hat{s}\,'(z_0)$. `notes/lemma_M_refuted.md`, §4. □

Proposition 5.5 (Bulge rule at $n = 7$; certified)**.** *For all* 124 *graphs, R^* was computed exactly (*51 *positive,* 73 *negative, none zero; every critical root simple and not shared with χ). A graph has a stable cell bounded above in t if and only if $R^* > 0$ and its fixed point is positive at β_c; this holds for exactly* 40 *graphs, and every bounded cell starts exactly at β_c. The* 11 *graphs with $R^* > 0$ that lose positivity before β_c do not bulge as fixed points. For E_0, $R^* = 1/4$ and $\beta_c = 2$.*[20]

At $n = 6$ no stable cell is bounded in t (Geneson's class has $R^* < 0$, and the limit interval of H_3 ends at a positivity root), so the $\varepsilon \to 0$ classification describes the six-node regions faithfully, while it understates the regions of 40 of the seven-node graphs and misses E_0 entirely.

5.6 The threshold at $n \le 7$

Theorem 5.6 (Threshold for $n \le 7$; certified, hand-verified)**.** *Let $\rho \approx 1.3247$ be the plastic number, the real root of $q^3 - q - 1$ (isolated in $(45/34, 4/3)$). Then $q_7^* = \rho$, and the supremum is not attained. Consequently, for every graph on at most seven nodes the target-free clique conjecture holds at every legal parameter with $q \ge \rho$, and for every $q \in (1, \rho)$ there is a legal parameter with that value of q at which it fails for a seven-node graph.*

Sketch. On every finite cell $q < \beta - 1$. On the 258 stable cells unbounded in t the supremum is $\beta_{\mathrm{hi}} - 1$ with β_{hi} a limit endpoint, and the largest is $1 + \rho$ (six graphs, whose limit interval ends at the positivity root $\beta^3 - 3\beta^2 + 2\beta - 1 = 0$, i.e. $q^3 - q - 1 = 0$, with the stability boundary further right). On the 40 bounded cells the outer isolating bound gives $\beta_{\mathrm{hi}} \le 16/7$, so $q < 9/7 < \rho$. Four of the six plastic-number graphs have limit intervals starting at $q = 1$, so the union of regions covers $(1, \rho)$. Combine with Theorem 4.4 and Corollary 2.7. An independent run of the exact threshold tool over all 298 stable cells returns matching upper and lower bounds with no undecided cell.[21] □

Geneson's bound gives $q \ge 5 - 2\varepsilon$ at $n = 7$; the true threshold is the plastic number. The clique-class family of Theorem 7.3 alone reaches only about 1.2747 at $n = 7$, and other graphs approach the supremum.

6 The $\varepsilon \to 0$ map at $n = 8$

The finite-ε pipeline of Section 5 does not scale to the 1,793,359,192,848 digraphs on eight nodes, but the $\varepsilon \to 0$ limit does. By Lemma 2.4, the full support is limit-stable at a legal β iff $w(\gamma) > 0$ and $\alpha(A|_\perp) < -\gamma$, and the spectral part can be screened by an integer Hurwitz computation on the exact limit polynomial at up to 1.8 million digraphs per second per process (1.5–1.7 million per second aggregate in production). We call a non-complete graph a *limit counterexample* if its full support is positive and limit-stable at some rational $\beta > 2$; by Lemma 2.4 its full support is then a stable non-clique fixed point at every legal point $(\varepsilon, \delta) = (1/(1+t), (\beta-1)/(1+t))$ with t large, so S_H is nonempty. This section reports the complete limit map at $n = 8$. Its status differs from the $n \le 7$ maps in two ways, stated here and repeated where they matter. It is a map of the *limit* only: counterexamples visible only at finite ε, of which Theorem 7.5 and Lemma 7.4 already provide eight-node examples, are outside it. And its 9,133 counterexamples have been verified by two independent machine campaigns but by no human hand check (Section 8).

[20] `certificates/n7_bulge_Rstar_exact.json`; `notes/bulge_exact/`.

[21] `certificates/qstar/qstar_n7_finite.json`; `certificates/qstar/qstar_n7limit.json`.

6.1 The restricted enumeration

By Lemma 3.6 a positive stable full support on eight nodes needs at least 29 edges, hence an underlying simple graph with at least 15 of the 28 possible undirected edges. There are 5,350 simple graphs on eight nodes with 15 to 28 edges and, by Burnside's lemma over $\mathrm{Aut}(G)$ for each of them, exactly 801,558,826,511 digraphs up to isomorphism with such an underlying graph and at least 29 edges. The remaining 991,800,366,337 digraphs on eight nodes (OEIS A000273 gives 1,793,359,192,848 in all) have at most 28 edges and are never positive and stable at any legal parameter, by the Lean-checked edge bound; they were not enumerated.[22]

Each underlying graph was enumerated once, its digraphs generated by nauty's `directg` from `geng` output, in 9 shards for the sparser graphs and 3,226 jobs for the denser ones; 201 graphs were split over several jobs, and every graph's output count was checked against its exact Burnside count. Two groups of graphs with large automorphism groups were enumerated by a different scheme, because the cost of `directg`'s isomorph rejection grows with the automorphism group and would have been days for them. *(i)* K_8 (575,009,339 digraphs, $|\mathrm{Aut}| = 40{,}320$): the isomorphism classes of digraphs with underlying graph G are in bijection with the pairs (a subgraph B of G up to $\mathrm{Aut}(G)$, an orientation of $G-B$ up to the stabiliser of B), where B is the set of bidirected pairs. For $G = K_8$ this replaces one enumeration with group S_8 by 12,345 enumerations with small groups, B ranging over the nonempty graphs on eight nodes; they were run as 40 batches whose per-B ledger sums to exactly 575,009,339. *(ii)* $K_8 - K_3$, $K_8 - K_{1,5}$ and $K_8 - (K_2 + K_{1,5})$, by the relative form of the same bijection implemented directly with least-element orbit representatives, in 14 parts with exact totals 1,179,317,832, 330,741,720 and 84,866,528. Both schemes re-enumerate the same isomorphism classes; neither is a prune. They were validated before use by canonical-form multiset equality against `directg` at $n = 4, 5, 6, 7$ (38, 570, 21,424 and 2,141,832 classes for K_n) and on further graphs, and after the run by re-enumerating $K_8 - e$ (5,339,129,738 digraphs, the richest graph) with the orbit scheme: its 143,266 canonical strict-survivor lines are exactly those of the 321 `directg` parts used in production. Four adversarial referees of the lemma, the code, the counts and the deployment found no defect.[23]

6.2 The accounting chain

Table 3 gives the chain, and Figure 2 shows it on a logarithmic scale. Stages 2–4 were run by an overflow-guarded C prefilter that is byte-identical to the $n = 7$ prefilter on all 1,540,944 six-node digraphs and on the 638,042 seven-node spectral survivors; it reads each digraph once, tests domination in the verbatim form of Lemma 3.5, and applies the Hurwitz test to the exact integer limit polynomial, returning three values. Only the 347,639 strict survivors (all Hurwitz determinants positive, i.e. $\alpha < -1/2$) were stored; the 417,603,634 "B" lines, for which a determinant vanishes before any negative one, have $\alpha \geq -1/2$ and are therefore never limit-stable at a legal β (at every $\beta > 2$, $-\gamma < -1/2 \leq \alpha$), but they were discarded at the shell level and are not available for any finite-ε question.[24] Stages 5–6 are the one-parameter engine of Section 3 at $t = \infty$ with replay, run on every stored line: the single run was interrupted by a container restart after 59,610 lines and finished as four shards of the unmodified script over disjoint subsets of 86,910, 86,910, 86,910 and 86,909 lines (2,312, 2,282, 2,310 and 2,229 non-complete stable graphs, K_8 in the second shard), merged by a checked script; the shard tallies sum

[22] `certificates/n8_limit_summary.json` (`scope`, `enumeration`); `certificates/n8_summary.json` (`chain.restricted_space`); `n8/n8_restricted_counts.json`; `edge_bound_ctln`^{L}, `notes/lean/CtlnCert/CtlnCert/EdgeBound.lean`.

[23] `notes/k8_decomposition.md` (lemma with proof, both forms); `notes/k8_decomposition_referees.json`; ledgers `n8/check_k8decomp.log`, `n8/check_orbit.log`; `n8/orbitenum/k8e_check/`; `certificates/n8_summary.json` (`chain.swaps`); `ANOMALIES.md`, entries of 27 and 28 October 2026.

[24] `n8/n8_prefilter_TOTAL.txt`; `n8/n8_accounting.json` (`ok = true`: 9 shards, 3,226 jobs, 201 split graphs, kept lines $S = 347{,}639$, $O = 0$); prefilter `n7/filter8.c`; runbook `n7/REPRODUCE_n8_limit.md`.

stage	condition (where proved)	removed	remaining
0	digraphs on eight nodes (OEIS A000273)		1,793,359,192,848
1	at most 28 edges (Lemma 3.6); not enumerated	991,800,366,337	801,558,826,511
2	a dominated node (Lemma 3.5)	687,860,834,577	113,697,991,934
3	a negative coefficient of the exact limit polynomial, or a negative Hurwitz determinant met before any vanishing one: $\alpha \geq -1/2$, never limit-stable (Lemma 2.4(b))	113,280,040,661	417,951,273
4	no negative coefficient, but a vanishing coefficient or a vanishing Hurwitz determinant met before any negative one (the "B" lines): $\alpha \geq -1/2$, never limit-stable; not stored	417,603,634	347,639
5	no $\beta > 2$ with $w(\gamma) > 0$ (Lemma 2.4(c))	331,427	16,212
6	positive and limit-stable at no common β (exact limit cells)	7,078	9,134

Table 3: Accounting chain of the limit map at $n = 8$. The 9,134 survivors are K_8 and the 9,133 limit counterexamples; 0 escalations and 0 overflow lines. Stages 2–4 are the C prefilter, stages 5–6 the exact one-parameter engine at $t = \infty$. Stage 1 counts are from the Burnside enumeration, stages 2–4 from the prefilter totals, stages 5–6 recounted from the per-graph records.

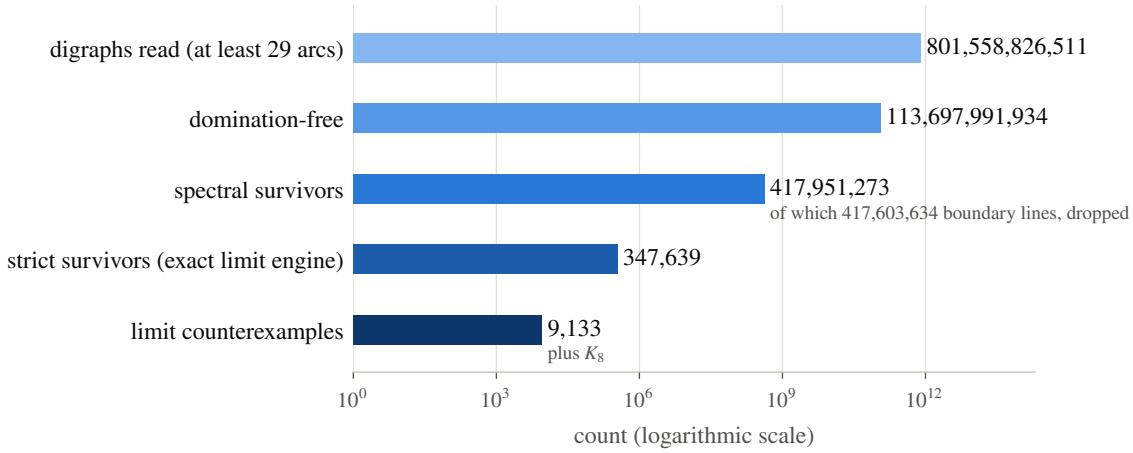


Figure 2: The accounting chain of the $n = 8$ limit map (Table 3) on a logarithmic scale: digraphs read (801,558,826,511), domination-free (113,697,991,934), spectral survivors (417,951,273, of which 417,603,634 boundary lines were dropped), strict survivors (347,639) and limit counterexamples (9,133, plus K_8). Bar length is $\log_{10}$ of the count. Counts from `n8/n8_prefilter_TOTAL.txt` and `n8/n8_limit_tally.json`.

to the merged tally and each of the 64,196 complete records of the interrupted run equals the shard record for the same graph. The limit stage took 10,776.6 CPU-seconds, about 31 ms per line.[25]

Theorem 6.1 ($n = 8$, limit map; certified, machine-verified, not hand-verified)**.** *Exactly* 9,133 *non-complete digraphs on eight nodes are limit counterexamples. Every other digraph on eight nodes is either excluded by a necessary condition valid at every legal parameter (stages 1–2 of Table 3) or certified not limit-stable at any legal β (stages 3–6). In particular the target-free clique conjecture fails for eight-node CTLNs at all sufficiently small ε on the ray $\delta = (\beta - 1)\varepsilon$ for every β in the stable interval of one of the*

[25] `n8/n8_limit_tally.json`; `n8/limit_sharding/shard\{0,1,2,3\}_tally.json`; `certificates/n8_summary.json` (`chain.limit_stage`, `counts`); script `n8/run_n8_limit.py`.

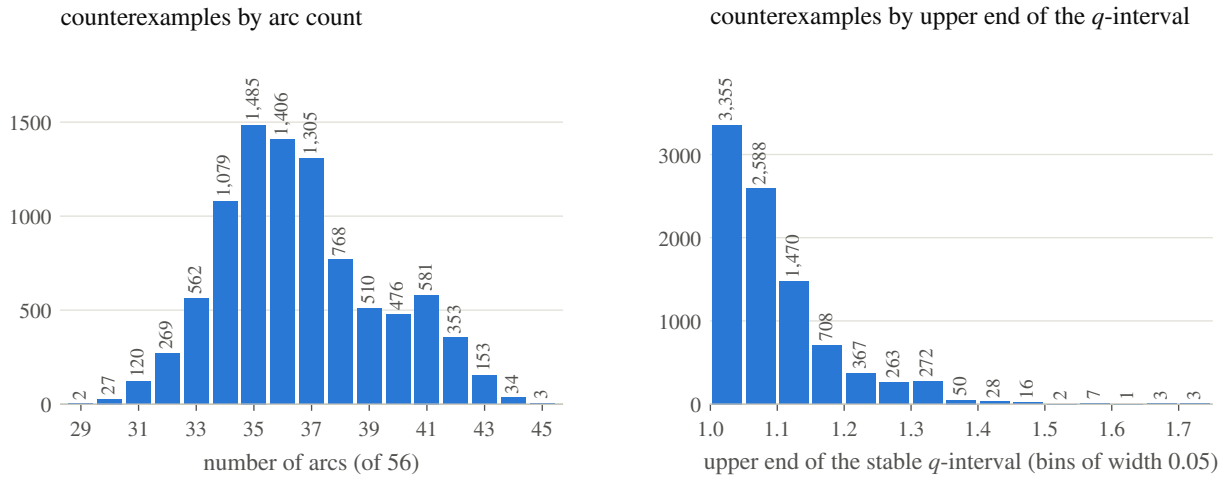


Figure 3: The 9,133 eight-node limit counterexamples by number of arcs (left; the counts of Table 4, recounted from the adjacency matrices in `n8/n8_limit_HITS.jsonl`) and by the upper end of their stable q-interval (right; bins of width 0.05 on $(1, 7/4]$, half-open on the left). A graph is placed in a bin by the certified upper bound of the refined isolating interval of its upper end (`certificates/n8_summary.json`, `rows`); no isolating interval straddles a bin boundary, and the counts are consistent with the histogram of Section 6.4 (coarser below 21/20, finer above). Counts are printed above the bars.

9,133 *graphs, and the tail of no legal ray is stable and positive for any other non-complete full support on eight nodes.*

The theorem says nothing about stability at finite ε outside those rays: the finite-ε map at $n = 8$ is not done: finite-ε-only counterexamples on eight nodes exist (15 distinct graphs satisfying Lemma 7.4, E_1 among them, all with $\alpha = -1/2$ and hence among the unstored B lines; at $n = 7$ there was exactly one such graph), and further ones may sit among the B lines, the 7,078 never_full lines or, unless Question 9.5 is answered, the lines with $\alpha > -1/2$. Every statement about $n = 8$ in this paper carries that scope.

6.3 The 9,133 graphs

All counts below are recomputed from the per-graph records and asserted against the limit tally and the prefilter totals.[26]

edges	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
graphs	2	27	120	269	562	1,079	1,485	1,406	1,305	768	510	476	581	353	153	34	3

Table 4: Edge counts of the 9,133 eight-node limit counterexamples (56 edges possible; the edge bound is ≥ 29).

Figure 3 shows the distribution of the edge counts and of the upper ends of the stable q-intervals.

[26] `n8/n8_limit_HITS.jsonl` (one record per graph: adjacency, exact stable interval with minimal polynomials and isolating intervals, four pointwise witnesses); `certificates/n8_summary.json`; `certificates/n8_counterexamples_classes.json`; `certificates/n8_candidates_legal_witnesses.json`; script `n8/aggregate_n8.py`.

- They are pairwise non-isomorphic (nauty canonical labelling), non-complete and domination-free; between 11 and 27 of the 56 edges are missing (Table 4). The underlying simple graph is complete for 6,037 of them; 2,895 have one non-adjacent pair, 198 two and 3 three. The number of bidirected pairs ranges from 2 to 17 of 28. The edge bound is *attained*: the two 29-edge graphs of Table 5 have 2 and 3 bidirected pairs against 1 and 2 non-adjacent pairs, the minimum Lemma 3.6 allows (at $n = 7$ the bound 22 was not attained; the sparsest counterexample had 23 edges).

- Each has exactly one stable β-interval, an open cell between consecutive critical roots. Its lower end is $\beta = 2$ (so $q \to 1$) for 8,162 graphs and a positivity root for 971; its upper end is a root of the Hurwitz determinant Δ (the spectral boundary β_c) for 7,191 graphs and a positivity root for 1,942. The upper ends have 4,298 distinct minimal polynomials, all irreducible, of degrees 1 (2 graphs), 3 (208), 4 (34), 5 (947), 6 (874), 7 (307), 10 (535), 12 (8), 15 (4,022) and 21 (2,196); the most frequent is again $q^5 + 2q^4 + q^3 - 2q^2 - 2q - 1$, the quintic of H_3 (408 graphs; 17 at $n = 7$), and the plastic cubic $q^3 - q - 1$ ends 178 intervals. The two degree-one ends are the rational value $q = 5/3$ ($\beta_c = 8/3$), discussed below. Every stored isolating interval contains exactly one root (Sturm), and the stored q-polynomial is the β-polynomial at $\beta = 1 + q$, for all 9,133.

- They form 6,260 classes of identical full-support data in the sense of Section 4 (same characteristic polynomial and in-walk counts): 4,939 singletons, 824 pairs, 2 triples, 374 classes of 4, 16 of 5, 3 of 6, one of 7, 84 of 8, 2 of 10, one of 12, 4 of 14, 8 of 16, one of 19 and one of 32. Within every class the stable interval, its minimal polynomials and the edge count are identical. 4,271 graphs carry a bidirected in-twin pair (5,202 pairs in all); all 29,116 in-twin switches (Proposition 4.3) of these graphs are again in the list, with identical intervals.

- *Lifts versus new graphs.* The 123 limit counterexamples on seven nodes have 858 distinct closed-twin blow-ups (one node doubled, Lemma 7.1); exactly 61 of them, from 52 parents, are in the list, and the other 797 are not limit counterexamples (prefilter: 690 dropped, 47 B, 60 never in the limit stage), and none is stable and positive at 4,782 sampled legal points inside its parent's interval. So 9,072 of the 9,133 are not single closed-twin lifts of a seven-node counterexample. Lemma 7.1 transfers the fixed-point data of the weighted quotient, not the region: 46 of the 61 lifts have an interval different from their parent's, one of them disjoint (`&G^z~Vwn?zBRw`). Conversely, 125 graphs in the list have a closed-twin pair whose contraction is *not* a counterexample (114 of these contractions are not even strict seven-node survivors). All 626 eight-node members of the clique-class catalogue of Section 7 (593 distinct graphs) are in the list, an independent recall check on the proved families (it cannot detect graphs outside them); F_2 of Theorem 7.3 is among them with its interval ending at the plastic number, as proved. The 15 graphs satisfying Lemma 7.4 (E_1 of Theorem 7.5 is one of them) have $\alpha = -1/2$ and are absent, as they must be.[27]

- *Witnesses.* Each record stores four pointwise witnesses on the ray through the midpoint of its interval, at $t = 10^2, \ldots, 10^5$; for 9,051 graphs all four are exact stable positive fixed points. As at $n = 7$, the $t = 10^2$ witness is often illegal ($q \leq 1$: 6,435 graphs), and for 37 graphs whose interval is narrower than 1/1000 in β the producer's fallback placed all four witnesses outside the interval (the records honestly store them as unstable). A separate file gives, for every graph, a rational β certified strictly inside its interval and a legal t at which the point is exactly stable and positive.[28]

[27] Lifts, switches and catalogue membership: `certificates/n8_summary.json` (`lifts`, `lemma_J_switch_closure_second_campaign`), `notes/verify_n8_2/completeness_twins/`, `notes/n8_structure_tools/families.json`, `analysis2.json`, `consistency_extra.json`.

[28] `certificates/n8_candidates_legal_witnesses.json` (field `witness_in_first_stable_cell`).

- *All large t.* For every graph, at a certified interior β, the exact set of $t > 0$ at which the full support is a legal stable positive fixed point is a single interval (t^*, ∞) with $t^* \le 10^4$; the largest t^* is an algebraic number in (31024237/3622, 3880171/453), about 8565.5, and the median is 23. For 8,388 of the 9,133 the binding constraint is legality itself: the full support is stable and positive at *every* legal t on that ray.[29]

graph	adjacency rows	edges	(B, O, N_n)	stable q	upper end
`&GM?WUZeEie^O`	00111000 00000110 00010110 01101110 01100001 10101010 10011001 11110100	29	$(3, 23, 2)$	$(1, q_c)$	Δ, $q_c \approx 1.0131$
`&GMBO]At]Xp]?`	00111000 00110100 00011110 00001011 01010111 10011001 11000101 11100000	29	$(2, 25, 1)$	$(1, q_c)$	Δ, $q_c \approx 1.0012$

Table 5: The two eight-node limit counterexamples attaining the edge bound (row i lists the j with $i \to j$; B, O, N_n as in Lemma 3.6). Both are rigid, have no twins of any kind, no node sending to all others, no smaller counterexample as an induced subgraph, and bidirected clique number 2. The upper ends are roots of degree-21 polynomials isolated in (64800179/63963136, 16200045/15990784) and (39760951/39714816, 106029203/105906176) by the first verification campaign.

6.4 The limit threshold at $n = 8$

Write $q_n^*(\infty)$ for the supremum of $q = \beta - 1$ over the limit-stable intervals of the limit counterexamples on n nodes. Since $q < \beta - 1$ on every ray and $q \to \beta - 1$ as $t \to \infty$, $q_n^* \ge q_n^*(\infty)$, with equality at $n = 6$ and $n = 7$ (Theorems 4.4 and 5.6: the suprema are approached as $t \to \infty$).

Theorem 6.2 (Limit threshold at $n = 8$; certified, machine-verified)**.** *$q_8^*(\infty)$ is the unique root of the irreducible polynomial*

$$13109q^{15} - 124967q^{14} + 538385q^{13} - 1420107q^{12} + 2727809q^{11} - 4550635q^{10} + 7298813q^9 - 10746735q^8 + 13594735q^7 - 15127269q^6 + 16652387q^5 - 18092785q^4 + 16098203q^3 - 9885513q^2 + 8301263q - 9045013$$

in the interval

$$[\,459224485/268435456,\ 2755346911/1610612736\,],$$

whose width is below 10^{-9}; $q_8^*(\infty) \approx 1.71074$. It is the common upper end of the intervals of the two graphs `&GMx|\Aw^qz]w` and `&GVrm]A{^pZ]w` (40 edges, the same class of identical full-support data, one bidirected in-twin switch apart), where the Hurwitz determinant Δ vanishes; every other upper end is certified strictly smaller. The supremum is not attained. Consequently $q_8^* \ge q_8^*(\infty) > \rho = q_7^*$.

Sketch. The upper end of each of the 9,133 intervals was refined by exact bisection to an isolating interval of width below 10^{-9} with a Sturm count of one; all but the two named graphs have their interval entirely below that of the two, and the two share the same irreducible polynomial with Sturm count one on the hull of their intervals. The records certify the open cell only; at the root a Hurwitz determinant vanishes, which is three-valued undecided and never counted as stable; at β_c itself $\alpha = -\gamma$, so the graph is not limit-stable there (Lemma 2.4). The refinement agrees with the independent refinement of the first verification campaign.[30] □

[29] `notes/verify_n8_2/interval_all_t/tally.json`.

[30] `certificates/n8_summary.json` (`q_star_limit`, `q_data`); `notes/verify_n8/v_structure/q_upper_ends_refined.json`.

The distribution of the 9,133 upper ends is: 37 in $(1, 1001/1000]$, 541 in $(1001/1000, 101/100]$, 2,777 in $(101/100, 21/20]$, 2,588 in $(21/20, 11/10]$, 2,178 in $(11/10, 6/5]$, 630 in $(6/5, 13/10]$, 322 in $(13/10, 7/5]$, 44 in $(7/5, 3/2]$, 9 in $(3/2, 8/5]$, 4 in $(8/5, 17/10]$ and 3 in $(17/10, 7/4]$; none equals one of these rationals and none exceeds 7/4. The smallest upper end belongs to `&GMBQMHtMH~]W`: a root of a degree-21 polynomial in (77079731/77070336, 1079116235/1078984704), about 1.000122; the graph is limit-stable at $\beta = 20001/10000$ ($q = 10001/10000$) and not at $\beta = 10001/5000$ ($q = 5001/5000$, Hurwitz determinant 6 negative).[31] Whether the union of the 9,133 limit intervals covers all of $(1, q_8^*(\infty))$, as the union of the seven-node regions covers $(1, \rho)$, is not recorded in the certificates; by Corollary 2.7 and Theorem 5.6 the conjecture fails at eight nodes for every $q \in (1, \rho)$ in any case.

The sixteen graphs with upper end above 3/2 form ten classes of identical full-support data. They are dense (37 to 42 edges, 9 to 14 bidirected pairs, 14 of them with no non-adjacent pair), rigid, without closed twins; ten end at the spectral boundary and six at a positivity root; only two contain a seven-node counterexample. The extremal pair has 12 bidirected pairs, no non-adjacent pair, bidirected clique number 4, trivial automorphism group and *no* node sending to all others. In the non-edge picture of Section 7, $\psi_D = z\,(8z^6 + 16z^5 + 34z^4 + 33z^3 + 30z^2 + 13z + 5)$, whose roots have minimum real part about -0.3689, so $\beta_c \approx 2.7107$; the positivity set is the whole half-line beyond the root ≈ 2.1537 of $b^4 - 3b^3 + 3b^2 - 3b + 1$ in $(2, 3)$, so the stable interval is $(2.1537\ldots, 2.7107\ldots)$ in β. The ceiling is thus set by the spectrum of the eight-node core itself, not by a sink class: its minimum real part -0.3689 lies to the *left* of the $k \to \infty$ limit -0.3516 of the sink-class family (Q) of Theorem 7.6, which has a seven-node core and tends to $q \approx 1.844$; family (Q) overtakes the eight-node record only for very large class size (q close to 9/5 needs $k \geq 286$). The record value 1709/1001 of Table 7 came from `&G^ZnRba\ba^w`, which is in the list (as `&GZptUbY\B}^w`, 39 edges) with refined upper end in (35849319/20971520, 401512373/234881024), about 1.7094. Two graphs, `&GMw|Vr]Rb~Nw` and `&GUo~EX~Yxz^w`, have a *rational* spectral end: $\psi_D = z\,(4z^2 + 3z + 1)(2z^4 + 2z^3 + 4z^2 + 2z + 1)$, the quadratic factor has real part exactly $-3/8$, so $\beta_c = 8/3$ and $q = 5/3$ exactly, not attained; their positivity set is the interval between the root ≈ 2.2056 of $b^3 - 2b^2 - 1$ and the root ≈ 2.7693 of $b^3 - 3b^2 + b - 1$.[32]

6.5 What the 9,133 share

The following statements were checked exactly on all 9,133 records; where a statement is proved, the proof is cited. The check files are listed in the footnote.[33] Figure 4 shows two of the graphs: the record graph of Theorem 6.2 and one of the two 29-arc graphs of Table 5.

- *Far from cliques.* The missing-edge digraph $D = J - I - A$ is never a matching, a star, a union of stars or a forest, and it always contains a directed cycle (proved: a D-source with a D-out-edge would make $u = (D^T + rI)^{-1}\mathbf{1}$ negative). The largest bidirected clique has at most $5 = n - 3$ nodes, attained by 61 graphs and never exceeded; no graph contains an induced K_6. The analogous bounds $n - 3$ hold and are attained at $n = 6$ (2 graphs) and $n = 7$ (16).

- *Universal properties.* Every graph is strongly connected, indeed has a Hamiltonian directed cycle; every graph contains an induced oriented 3-cycle, the only induced subgraph on 3 to 7 nodes common to all 9,133 (the census found 15, 169, 3,040, 34,521 and 48,108 distinct induced

[31] `notes/n8_structure_tools/check_min_q.json`. Two summary files, `certificates/n8_limit_summary.json` (`q_upper_end_min_approx`) and `notes/verify_n8/SYNTHESIS.json`, print 1.000276 for this minimum; the per-graph refinement file of the same campaign, the aggregation certificate and the exact check above all give the value quoted here. The display value is not used by any certificate.

[32] `notes/n8_structure.md`, §1.8; `notes/n8_structure_tools/special_sets.json`.

[33] `notes/n8_structure.md` with the tools and outputs in `notes/n8_structure_tools/` (`structure_summary.json`, `motifs.json`, `heredity.json`, `families.json`, `analysis2.json`, `positivity_check.json`, `discriminate.json`, `ham_never.json`, `special_sets.json`).

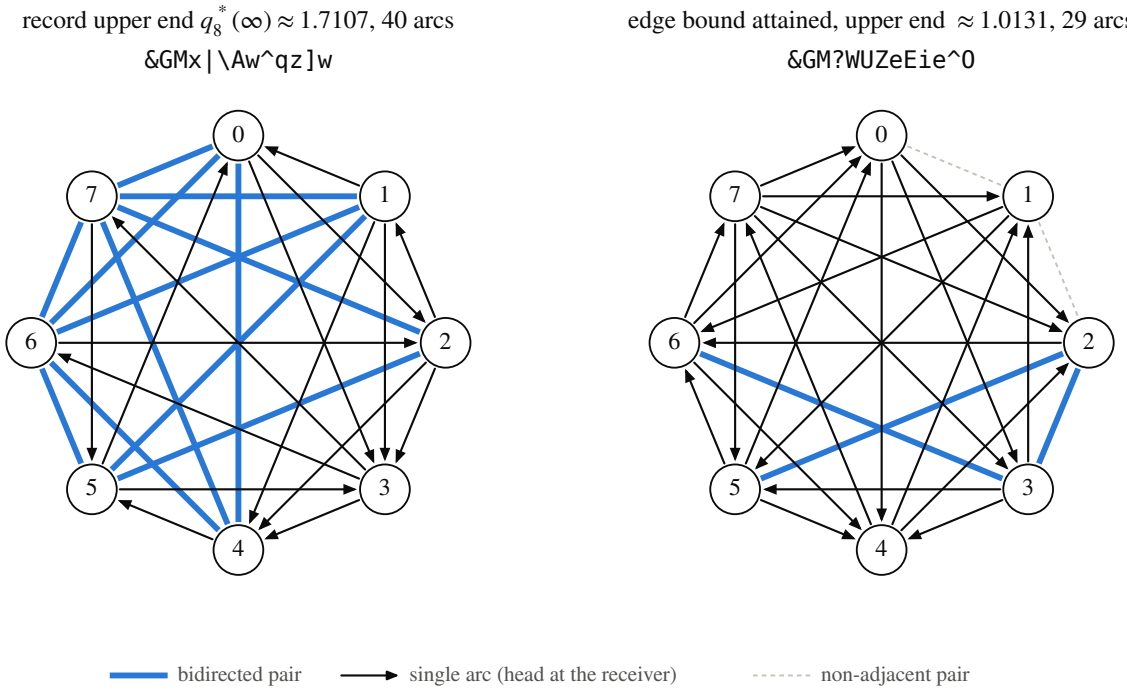


Figure 4: Two eight-node limit counterexamples, drawn as in Figure 1 (vertices in digraph6 order, clockwise from the top). Left: `&GMx|\Aw^qz]w`, 40 arcs, 12 bidirected pairs and no non-adjacent pair, whose stable interval has the largest upper end, $q_8^*(\infty)$ (Theorem 6.2; its twin switch `&GVrm]A{^pZ]w` has the same interval). Right: `&GM?WUZeEie^O`, one of the two graphs attaining the edge bound, 29 arcs, 3 bidirected pairs and 2 non-adjacent pairs (Table 5). Adjacency from `n8/n8_limit_HITS.jsonl`; decimals in the panel titles are display values of the certified upper ends.

subgraphs on $3, \ldots, 7$ nodes); a node that receives from all others also sends to all others (proved; 131 graphs are joins $H \vee K_1$ or $H \vee K_2$); no two non-adjacent nodes are in-twins (Lemma 3.8). None of these singles out counterexamples: the 338,505 strict survivors of stages 2–4 that are *not* limit counterexamples satisfy all of them, except that 18 of them are not strongly connected (all 18 fail positivity). What separates the 9,133 is the joint condition, positivity and $\alpha < -\gamma$ at a common β, and no graph-local feature we tested captures it.

- *Properties that fail.* 5,276 graphs have no node sending to all others (3,185 have one, 562 two, 110 three); 3,096 have a non-adjacent pair; 971 have an interval not starting at $\beta = 2$; 1,942 have an interval ending at a positivity root; 188 have a nontrivial automorphism group (186 of them because of closed twins). For the 3,857 graphs with a node sending to all others, the sink-class criterion of Lemma 7.2(X2) holds at a certified interior β in every case (w of the core positive and every sink's in-set of weight above 1); in 1,972 of them, deleting any one such node leaves a graph that is not a seven-node limit counterexample (1,970 leave no seven-node counterexample at all; in two, a sink deletion gives `&FNFL~[udo?`).

- *Heredity fails.* 6,720 graphs contain no six- or seven-node counterexample as an induced subgraph (2,323 contain a seven-node one, and all 123 occur; 308 contain a six-node one). For 5,902 graphs no single-node deletion is even a strict seven-node survivor (domination-free with $\alpha < -1/2$). At $n = 7$, 102 of the 123 limit counterexamples contained no six-node one. At $n = 7$ and $n = 8$ most counterexamples are not one-node extensions of smaller ones.

- *Mechanism of the interval.* The exact positivity set on $(2, \infty)$ is one interval for 8,984 graphs and two for 149, unbounded above for 2,133; the stored stable interval lies in one component of it for all 9,133 (0 inconsistencies). The lower end is never spectral: when positivity holds down to $\beta = 2$, so does limit stability, as Lemma 2.4(b) requires. The six combinations (lower end 2 or positivity root; upper end spectral with bounded positivity, spectral with unbounded positivity, or positivity root) occur 4,690, 1,974, 1,498, 504, 23 and 444 times.

Conjecture 6.3 (Structure of limit counterexamples)**.** Let G be a non-complete digraph on $n \geq 3$ nodes whose full support is limit-stable at some $\beta > 2$. Then (S1) G has no bidirected clique on $n-2$ nodes, equivalently its missing edges cannot be covered by two nodes; (S2) G is strongly connected, indeed Hamiltonian; (S3) G contains an induced oriented 3-cycle.

Evidence: the complete limit maps at $n = 6, 7, 8$, where (S1) is attained at every n; every one of the 631 proved infinite families of Section 7 satisfies (S1) for every class size (with equality for all k in 38 of them, so $n-3$ cannot be lowered); 104 certified instances from $n = 7$ to $n = 73$ satisfy (S1)–(S3); and an exhaustive test of (S1) at $n = 9$: every digraph containing a bidirected K_7 is a multiset of 7 node patterns relative to the two outside nodes plus their mutual relation, 682,176 configurations in all, of which 600,440 have a dominated node, 81,735 fail the strict spectral test and exactly one survives, K_9. So (S1) holds for all $n \leq 9$ (for limit counterexamples, as the conjecture is stated). The same test at $n = 3, \ldots, 8$ leaves only K_n. Since a bidirected K_{n-2} forces D to have rank at most 4, (S1) is a statement about a finite-type integer family, of the kind the Hurwitz-in-k arguments of Theorems 7.3 and 7.8 handle.[34]

6.6 The threshold picture for $n = 6, 7, 8$

n	limit c.e.	all c.e.	$q_n^*(\infty)$	q_n^*	Geneson's bound $n-2$
≤ 5	0	0	–	–	
6	3	3	sextic root ≈ 1.1664	$= q_6^*(\infty)$	4
7	123	124	$\rho \approx 1.3247$	$= \rho$	5
8	9,133	$\geq$ 9,148, not mapped	degree-15 root ≈ 1.7107	$\geq q_8^*(\infty)$	6

Table 6: Counterexamples (c.e.: non-complete graphs with a stable positive full support) and thresholds. The $n = 8$ row is the limit map only, machine-verified and not hand-verified (its "all c.e." entry counts the 9,133 limit counterexamples plus the 15 known finite-ε-only graphs of Lemma 7.4); the $n \leq 7$ rows are complete finite-ε maps, hand-verified. Values of q^* are display approximations of the exact algebraic numbers in Theorems 4.4, 5.6 and 6.2.

Table 6 collects the thresholds and Figure 5 shows the stable q-intervals of the limit counterexamples for $n = 6, 7, 8$ on one axis. The counts grow by factors of about 41 and 74, the share of primitive counterexamples (containing no smaller one) grows from 3/3 to 102/123 to 6,720/9,133, and the threshold grows from 1.1664 to 1.3247 to at least 1.7107, against Geneson's $n-2$. At $n = 8$ the record is set by a sink-free core, without any sink class; the sink-class families of Theorem 7.6 have larger limits (-0.3516 for Q, $-1/4$ for Q2) but reach them only for large class sizes; the growth of q_n^* at small n comes from new cores, while the asymptotic growth of Theorem 7.7 comes from sink classes. A hand-check kit with six fully worked eight-node cases, in exact fractions, is provided for any reader who wants to verify a sample by hand.[35]

[34] `notes/n8_structure_tools/family_test.json`; `clique_class_test_n\{3,...,9\}.json`; `clique_class_control.json`.

[35] `notes/verify_n8_2/handcheck_kit/HANDCHECK.md`.

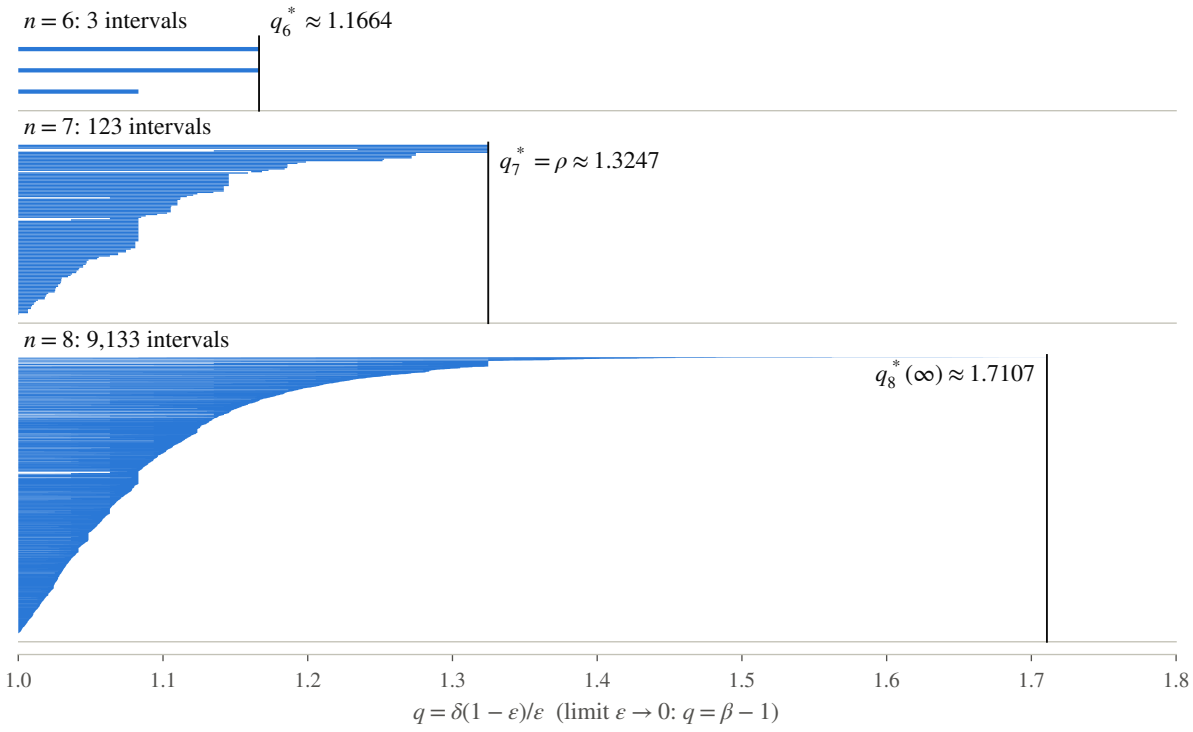


Figure 5: The threshold picture in the limit $\varepsilon \to 0$ ($q = \beta - 1$). For $n = 6$, 7 and 8 the stable q-intervals of the limit counterexamples (3, 123 and 9,133 intervals, from `certificates/n6_summary.json`, `certificates/COUNTEREXAMPLE_CANDIDATES_n7_limit.jsonl` and `certificates/n8_summary.json` with `n8/n8_limit_HITS.jsonl`) are drawn as horizontal segments, one row per graph, sorted by upper end (ties by lower end) from the bottom up, on a common q axis. For $n = 8$ the 9,133 rows are drawn as their envelope, one horizontal band per row between its lower and upper end, which is what the dense stack of rows of sub-point thickness would show; the sixteen rows ending above $3/2$ are too thin to see. White gaps inside a stack are rows whose interval starts above $q = 1$ (lower end a root rather than $\beta = 2$): 6 of the 123 rows at $n = 7$ and 971 of the 9,133 at $n = 8$, grouped because rows with equal upper ends are sub-sorted by lower end. Each segment runs from the upper bound of the isolating interval of its lower end (from $q = 1$ when the interval starts at $\beta = 2$) to the lower bound of the isolating interval of its upper end, so nothing is drawn longer than certified; the coarse stored isolating intervals at $n = 6, 7$ were first narrowed to width below 10^{-9} by exact bisection of the stored minimal polynomial, and every endpoint used is recorded in `notes/paper/figures/figures_data.json`. Vertical lines mark q_6^*, $q_7^* = \rho$ and $q_8^*(\infty)$ (Theorems 4.4, 5.6 and 6.2) at the midpoints of their isolating intervals, narrowed by the same exact bisection to width below 10^{-9} so that each line coincides with the drawn end of the record rows; the decimals are display values.

7 Infinite families

The theorems in this section are proved for every member of a family; tables and individual instances are certified only. Instances on $n \geq 8$ nodes are certified by the pointwise exact verifier but have *not* been hand-verified, and are reported as candidates in that sense. The proofs use the base graphs only as explicit matrices. Theorems 7.3, 7.5, 7.6 and 7.7 rest on Lemma 2.4(b),(c) (Lemmas I.3 and I.4 of the notes: transfer from the limit to finite t, and t-free positivity); Theorem 7.8 uses only part (c).

7.1 The non-edge picture and blow-ups

Let $D := J - I - A$ be the sender-first non-edge matrix ($D = N^T$ for the full support) and $\psi_D(z) := \mathbf{1}^T \operatorname{adj}(zI - D)\mathbf{1} = m\,\chi_{D|_\perp}(z)$. Since $A|_\perp = -I - D|_\perp$, Lemma 2.4 reads: *the full support is limit-stable at β iff every root of ψ_D has real part $> -1/\beta$*; and the fixed point is positive iff $u := (D^T + \beta^{-1}I)^{-1}\mathbf{1} > 0$ and $\mathbf{1}^T u > 1$, in which case $w = u/(\mathbf{1}^T u - 1)$.

Lemma 7.1 (Closed-twin blow-up; proved)**.** *Let H have m nodes and let $\mu \in \mathbb{Z}^m_{>0}$, $M = \operatorname{diag}(\mu)$. Let H^μ replace node a by a bidirected clique of μ_a nodes, with every node of class a sending to every node of class $b \neq a$ iff $a \to b$ in H; $n = \sum \mu_a$. Then*

$$n\,\chi_{D'|_\perp}(z) = z^{n-m}\,\mu^T \operatorname{adj}(zI - DM)\mathbf{1},$$

and for every γ with $A'^T + \gamma I$ nonsingular the positivity vector of H^μ is constant on classes, $w' = Ey$ with $\big((A^T + I)M + (\gamma - 1)I\big)y = \mathbf{1}$, where E is the class-indicator matrix.

Sketch. $D' = EDE^T$ and $E^T E = M$; push-through $(zI - EDE^T)^{-1}E = E(zI - DM)^{-1}$ and Sylvester's identity $\det(zI_n - EDE^T) = z^{n-m}\det(zI_m - DM)$.[36] □

For a base G, in-set S and out-set T, let $G[S, T; k]$ add a bidirected clique V of k nodes, with $i \to v$ for all $v \in V$ iff $S_i = 1$ and $v \to i$ iff $T_i = 1$; put $b = \mathbf{1} - S$, $c = \mathbf{1} - T$.

Lemma 7.2 (Clique-class extension; proved)**.** *(X1) $n\,\chi_{D'|_\perp} = z^{k-1}\Phi_k$ with $\Phi_k(z) = z\,\psi_D(z) + k\,\chi_L(z)$, $L := D - b\mathbf{1}^T - \mathbf{1}c^T$. (X2) If T is all of G, then for every $k \geq 1$ and rational $\beta > 1$ the full support of $G[S, T; k]$ is positive iff $w_G(\beta) > 0$ and $\sigma_S(\beta) := \sum_{i \in S} w_{G,i}(\beta) > 1$; this does not depend on k or t. (X3) As $k \to \infty$ the moving spectrum converges to $\operatorname{spec}(L)$. With several sink classes of sizes k_j and in-sets S_j, the same holds with L replaced by $D - \bar{b}\mathbf{1}^T$, $\bar{b}$ the size-weighted mean of the b_j, and positivity holds iff $w_G > 0$ and $\sigma_{S_j} > 1$ for every j.*

7.2 Counterexamples on every $n \geq 6$

Let G_1 be Geneson's graph in the paper's labelling (rows `011011 101011 110101 111000 011000 000110`), $S = (1, 1, 1, 1, 1, 0)$, and $F_k := G_1[S, \text{all}; k]$, $n = 6 + k$.

Theorem 7.3 (Clique-class family; proved, internally reviewed)**.** *For every $k \geq 0$, $\alpha(F_k|_\perp) < -1/2$, and F_k is positive at every $\beta \in (2, b_p)$, where $b_p - 1 = \rho$ is the plastic number. Hence for each k and each rational $\beta \in (2, \min(b_p, \beta_c(k)))$ the full support of F_k is a stable positive non-clique fixed point at $(\varepsilon, \delta) = (1/(1 + t), (\beta - 1)/(1 + t))$ for all large t. In particular the target-free clique conjecture fails for every $n \geq 6$.*

Sketch. By Lemma 7.2, $\Phi_k = z^2 R_k$ with $R_k = (6+k)z^4 + (11+k)z^3 + (16+k)z^2 + 10z + 3$. The shifted polynomial $16R_k(-\frac{1}{2} - x) = 16(k+6)x^4 + 16(k+1)x^3 + 8(2k+17)x^2 + 12(k+1)x + (3k+16)$ has Hurwitz determinants $16(k+1)$, $64(k+1)(k+16)$, $8192(k+1)^2$ and $8192(k+1)^2(3k+16)$, all positive for $k > -1$. Positivity: the components of $\operatorname{adj}(\beta A^T + (\beta - 1)I)\mathbf{1}$ and $\sigma_S - 1$ are explicit polynomials; exact root counting on $[2, 93/40]$ shows a single sign change, at the root b_p of $b^3 - 3b^2 + 2b - 1$.[37] □

The limit supremum of q along the family is $\min(b_p, \beta_c(k)) - 1$: about 1.1664 ($k = 0$), 1.2747 ($k = 1$), the plastic number 1.3247 for $k = 2, \ldots, 8$ ($n = 8, \ldots, 14$), then decreasing to 1 (since $\chi_L = z^4(z^2 + z + 1)$ has roots on $\operatorname{Re} z = -1/2$). The same construction proves 9 families on the three six-node graphs and 617 families on 117 of the 124 seven-node graphs, each for every $k \geq 1$, by exact real-root isolation in k; an independent referee re-proved all 626 for every real $k \geq 1$.[38]

[36] `notes/research_wf3/theory/notes_theory.md`, §2.

[37] `notes/research_wf3/theory/family_F_certificate.json` (script `family_F_proof.py`); witnesses for $k = 1, \ldots, 6, 8, 10$ ($n = 7, \ldots, 16$) at $(\varepsilon, \delta) = (1/21, 11/210)$ or $(1/21, 11/175)$.

[38] `notes/research_wf3/theory/families_catalogue_n6.json`, `families_catalogue_n7.json`.

7.3 Finite-ε-only counterexamples on every $n \geq 7$

Lemma 7.4 (Boundary mechanism; proved). *Suppose (i) ψ has a conjugate pair of simple roots $z_0, \bar{z}_0$ with $\operatorname{Re} z_0 = 1/2$ and $\chi(z_0) \neq 0$; (ii) every other root of ψ has real part $> 1/2$; (iii) $R := \operatorname{Re}\big(\chi(z_0)/\psi'(z_0)\big) > 1/8$; (iv) $w(2) = (A^T + \frac{1}{2}I)^{-1}\mathbf{1} > 0$. Then $\alpha = -1/2$, so the graph is never limit-stable at a legal β, yet its full support is a stable positive fixed point at legal points with ε arbitrarily small (and $\beta \to 2^+$).*

Sketch. By Theorem 5.4, $\nu_A(c) = \frac{1}{2} + R/c + O(c^{-2})$, while the legality boundary is $c - \sqrt{c^2 - c} = \frac{1}{2} + \frac{1}{8c} + O(c^{-2})$. If $R > 1/8$ there is room between them for all large c; choose a rational γ in between. Positivity holds near $\beta = 2$ by (iv) and is t-free. □

Let $H =$ `&FNDL~[uds?` (one of the 123) and let E_a be its blow-up with class sizes $(1, 1, 2, a, 1, 1, 1)$, $n = 7 + a$; E_0 is `&FNFL~[udo?`.

Theorem 7.5 (Finite-ε-only family; proved, internally reviewed). *For every $a \geq 0$, E_a is not a clique, $\alpha(E_a|_\perp) = -1/2$ exactly, and the full support of E_a is a stable positive fixed point at legal points with ε arbitrarily small. Thus for every $n \geq 7$ there is a counterexample on n nodes that is invisible in the $\varepsilon \to 0$ limit.*

Sketch. Lemma 7.1 gives $\Phi = z(z^2 + z + 1)\big((a + 7)z(z^2 + z + 1) + 2\big)$ identically in a (non-edge picture); the cubic factor has all roots in $\operatorname{Re} z > -1/2$ iff $a + 7 > 16/3$; in the adjacency picture, at $z_0 = \frac{1}{2} - \frac{\sqrt{3}}{2}i$ one has $1/\hat{s}'(z_0) = \frac{1}{4} - \frac{\sqrt{3}}{12}i$, so $R^* = 1/4 > 1/8$ for every a; and the class vector at $\beta = 2$ is $(6, 2, 2, 18, 6, 2, 10)/(18a + 17) > 0$. Apply Lemma 7.4. Instances $a = 0, \ldots, 5$ are certified at $t = 20$, $\beta = 821/400$, $(\varepsilon, \delta) = (1/21, 421/8400)$, $q = 421/420$.[39] □

The two-parameter engine applied to E_1 returns one stable cell, over the same β-cell and with the upper t-end at $\beta = 53/26$ in the same isolating bracket $(53, 54)$ as for E_0.[40] The two regions are not identical: at $\beta = 53/26$, $t = 107/2$ the full support of E_0 is stable and that of E_1 is not (Hurwitz determinant 7 negative), so the upper end of the window depends on a. The cube-root factor arises from the root locus of the class size: for the blow-up family of H_3 one has $\Phi_k(\omega) = (k - 1)\omega^2$ at a primitive cube root of unity ω, so the locus passes through ω exactly at class size 2. A float-steered scan of all one-node extensions of the 124 seven-node graphs, followed by exact triage of the 9,808 extensions it flagged (an eigenvalue of $A|_\perp$ near a primitive cube root of unity), found 15 eight-node graphs that satisfy every hypothesis of Lemma 7.4, each with an exact legal witness; extensions not flagged by the float prefilter were not examined exactly.[41]

7.4 Growing thresholds

Theorem 7.6 (Families Q, Q2, Y; proved, internally reviewed). *(Q) For the core `&F^S{^ReL|?` with one sink class of size k ($n = 7 + k$), every member is positive at every $\beta > 2$, and is limit-stable at $\beta = 5/2$, 27/10, 14/5 for all $k \geq 25$, 79, 286 respectively. So for every $n \geq 32$ (86, 293) there are counterexamples on n nodes with q arbitrarily close to 3/2 (17/10, 9/5); along the family the limit value of q tends to $1.84416\ldots$. (Q2) For the core `&F^U|m^QTq?` with two sink classes of sizes k and $3k$ ($n = 7 + 4k$), limit polynomial $z^3(2z^2 + z + 2)^2/4$, every member is positive at every $\beta > 2$, and is limit-stable at $\beta = 5/2$, 3, 7/2, 19/5 for all $k \geq 1$, 3, 10, 33. (Y) For the eight-node core `&G^ZZZr|KjqTo` with two sink classes of sizes $4k$ and k ($n = 8 + 5k$), every member is positive at every $\beta > 2$, and is limit-stable at $\beta = 3$, 7/2, 4, 9/2, 24/5 for all $k \geq 13$, 24, 46, 117, 363; the family ceiling is $\beta \approx 4.9607$.*

[39] `notes/research_wf3/theory/family_E_certificate_part1.json`, `family_E_witnesses.json`.

[40] `certificates/n8_instances_region2d.json`.

[41] `notes/research_wf3/theory/cube_n8_witnesses.json`.

Each threshold is proved sharp by exact real-root isolation in k of the Hurwitz determinants of the shifted limit polynomial. The eight-node core of (Y) is not among the hand-verified graphs, and it is not itself in the list of Theorem 6.1 (it needs its sink classes).[42] Certified instances include $n = 32$ at $q = 15000/10001$ (Q), $n = 19$ at $q = 20000/10001$ and $n = 47$ at $q = 25000/10001$ (Q2), and $n = 73$ at $q = 20000/10001$ (Y). Exact lower bounds at small n found by search and certified pointwise are listed in Table 7; the limit map at $n = 8$ raises its first row to $q_8^* \geq q_8^*(\infty) \approx 1.7107$ (Theorem 6.2).

n	graph	(ε, δ)	q (exact)
8	`&G^ZnRba\ba^w`	(1/1001, 1709/1001000)	$1709/1001 \approx 1.7073$
9	`&H^\us{\HrEa^|~_`	(1/1001, 17691/10010000)	$17691/10010 \approx 1.7673$
10	`&I^Qz|NbkfuNa^}~|tw`	(1/10001, 9209/50005000)	$18418/10001 \approx 1.8416$
11	`&J^R\~R{\a^k^a^~^~]n^~?`	(1/100001, 861/45455000)	$17220/9091 \approx 1.8942$

Table 7: Certified lower bounds $q_n^* \geq q$ (not hand-verified).

7.5 $q_n^* \to \infty$

A fixed core with sink classes cannot push the limit threshold past the number of core nodes: the limit matrix $D - \bar{b}\mathbf{1}^T$ has trace ≤ -1 when no class is a join. The way around this is a family of cores whose limit polynomial can be prescribed.

Theorem 7.7 (Unbounded thresholds; proved, internally reviewed)**.** *For every integer $N \geq 3$ there is $n_0(N)$ such that for every $n \geq n_0(N)$ some non-complete n-node digraph has a stable positive full-support fixed point at the legal points $(\varepsilon, \delta) = (1/(1+t), (N-1)/(1+t))$ for all large t. Hence $q_n^* \geq N-1$ for $n \geq n_0(N)$, and $q_n^* \to \infty$.*

Sketch. Arms forest. In the non-edge digraph D, take a directed 3-cycle $c_0 \to c_1 \to c_2 \to c_0$ and attach "arms" $c_2 \to a \to z_a \to$ (target), so that every node other than c_2 has exactly one D-out-neighbour and reaches c_2 (an in-tree rooted at c_2), while c_2 sends to c_0 and to the first node of each arm. By the matrix determinant lemma (Lemma T of the notes), $\chi_{D-e_{c_2}\mathbf{1}^T}(z) = z^{m-d}p(z)$ for any $p = z^d + z^{d-1} + c_2 z^{d-2} + \cdots + c_d$ with integers $c_2 \geq 1$, $c_j \geq 0$ satisfying a mild forest constraint; the core has $m = 2p(1) - 3$ nodes. Lemma Q (Sherman–Morrison) gives ψ_D in closed form and Lemma P gives positivity in closed form. A single sink class receiving from every core node except c_2 has $\Phi_k = z^{m-d}(G + kp)$ with G fixed. *Target polynomial.* With $s = 1/(2N)$ and an integer W, round the coefficients of $\prod_{k=1}^{N}\big((z+s)^2 + W^2k^2\big)$ (except the first two) to the nearest integer. By Rouché (Lemma S), for $W > W_0(N) = 2N\,4^N(N+1)^{2N-2}$ all roots of the rounded polynomial $\tilde{p}$ satisfy $-3/(4N) < \operatorname{Re} z < -1/(4N)$, and all its coefficients from c_2 on are ≥ 1. *Positivity* at $\beta = N$ holds for large W because the top coefficient dominates. *Class size.* $\Phi_k/(m+k) \to \chi_L = z^{m-d}\tilde{p}$, so by root continuity there is k_0 with all roots of Φ_k in $\operatorname{Re} z > -1/N$ for $k \geq k_0$. Lemma 2.4 at the rational $\beta = N$ finishes, with $n_0 = m + k_0$.[43] □

The bound $n_0(N)$ is super-exponential, and k_0 is not explicit in general. Exact reduced certificates give $n_0 = 1{,}812$, 6,214,995 and 1,590,755,528 for $N = 3, 4, 5$; a 65-node core with p-coefficients $(1, 1, 11, 7, 10, 3, 1)$ gives $q_n^* \geq 3$, 4, 9/2, 23/5 for all $n \geq 219$, 657, 2,475, 4,909. Two explicit graphs are certified directly by the pointwise verifier without using any lemma: $n = 28$ at $q = 20000/10001$ and $n = 41$ at $q = 22500/10001$; at $k_0 - 1$ the explicit limit test fails with a negative Hurwitz determinant, so the reduced thresholds are sharp there.[44] The only known upper bound remains $q_n^* \lesssim n-2$ from (2).

[42] `notes/research_wf3/theory/family_Q_certificate.json`, `family_Q2_certificate.json`, `family_Y_certificate.json`.

[43] Theorem U, `notes/theorem_U/`; referee record `notes/theorem_U_workflow.json`.

[44] `certificates/theorem_U_explicit_instances.json`.

7.6 Unbounded support size at one parameter point

Theorem 7.3 uses a k-dependent schedule $\beta \to 2$. At a fixed legal point the family F_k eventually dies: at $(\varepsilon, \delta) = (1/21, 11/210)$ it is stable exactly for $k = 0, \ldots, 40$ and unstable for every $k \geq 41$, and in fact at every fixed legal point for large k, because χ_L has the factor $z^2 + z + 1$ with roots on $\operatorname{Re} z = -1/2$. A different core avoids this.

Let $R =$ `&F^U|m^QTq?` (one of the 124; rows `0111110 1011011 1101101 1100111 1101001 0010101 1100100`), $S = (1, 1, 0, 1, 1, 1, 1)$ and $R_k := R[S, \text{all}; k]$, $n = 7 + k$.

Theorem 7.8 (One point, every size; proved, internally reviewed)**.** *At $(\varepsilon, \delta) = (1/21, 11/210)$, i.e. $t = 20$, $\beta = 21/10$, $q = 22/21$, the full support of R_k is a stable positive non-clique fixed point for every $k \geq 0$. Consequently, at this point a non-complete n-node digraph with a stable positive full-support fixed point exists if and only if $n \geq 6$.*

Sketch. In the non-edge picture, (3) holds iff every eigenvalue of $D' + aJ$, $a = t/\beta$, has real part $> -1/\beta$, and for sink-class families $\det(zI - aJ - D') = z^{p-1}\Psi(z)$ with $\Psi = z\det(zI - D - aJ) - a\sum_j k_j \det(zI - D + b_j \mathbf{1}^T)$. Here $\Psi_k = z^3 g_k$ with $g_k(z) = z^5 - 3z^2 - 2z - 1 - a[(k+7)z^4 + (k+12)z^3 + (3k+21)z^2 + (k+13)z + (k+5)]$. At $a = 200/21$, after the shift $z = -10/21 - x$, every Hurwitz determinant is a polynomial in k with only positive coefficients (for example $21\Delta_1 = 200k + 1450$), hence positive for all $k \geq 0$. Positivity: $w_R(21/10) > 0$ and $\sigma_S = 381654/295819 > 1$, so Lemma 7.2(X2) applies for every k. The "only if" part is Theorem 4.1; $n = 6$ is G_1.[45] □

An independent second proof (equitable-partition quotient, separate exact code) confirms the result; members up to $n = 107$ were certified pointwise. Also proved in the notes (the referee spot-checked the first statement at 28 exact legal points and re-proved the self-join statement independently): R_k is an SFP at *every* legal point with $\beta \leq 43/20$, for every $k \geq 0$; every legal point with $\beta < 4.9607$ carries an infinite family (via Theorem 7.6(Y)); and, conditionally on the lemmas of Theorem 7.7, every legal point does, with an exact instance at $(\varepsilon, \delta) = (1/2, 2)$ for all $k \geq 1581$. The k-fold self-join of R ($n = 7k$), which is not a blow-up, is an SFP at every legal point with $\beta \leq 21/10$ for every $k \geq 1$.

Proposition 7.9 (A refuted conjecture; certified)**.** *It is not true that positivity of $(D^T + rI)^{-1}\mathbf{1}$ for some $r \in (0, 1/2)$ forces D to have an eigenvalue with real part $\leq -1/2$. For example, the non-edge matrix of `&F^U|m^QTq?` has every eigenvalue with real part $> -10/21$ while $(D^T + \frac{10}{21}I)^{-1}\mathbf{1} > 0$; and 33 of the 54 catalogue entries with S equal to all nodes, e.g. the base `&FMAlW\d\y?` with $\chi_D = z^7 - z^5 - 7z^4 - 10z^3 - 9z^2 - 4z - 1$, have every eigenvalue of D in $\operatorname{Re} z > -1/2$ and a positive vector at $r = 49/100$. In particular the joins $G \vee K_k$ of such bases are limit-stable for every $k \geq 1$.*

(This was "Conjecture P" of the project notes; see Question 9.5.)

8 Verification and reproducibility

8.1 Certificates

Every number in Sections 4–7 is read from a certificate file that can be regenerated by a named script (Appendix A). Values are stored exactly: integers, rationals, and algebraic numbers as an irreducible integer minimal polynomial together with a rational isolating interval; decimals appear only as labelled approximations. A region certificate stores, for each cell, the isolating data of its walls, an exact sample, the verdict with its reason (for example "Hurwitz determinant 6 negative"), and the outcome of the pointwise cross-check. The "never" records of the large sweeps store critical values and cell counts rather than full decompositions, so verifying them means re-running the committed engine; the records of all stable graphs are self-contained. Bulk files carry SHA-256 checksums.

[45] Theorem A, `notes/theorem_A_fixed/`; `certificates/theorem_A_fixed_parameter_instances.json` ($k = 0, \ldots, 15$); referee record `notes/theorem_A_fixed_workflow.json`.

8.2 Independent audits

The engine and the $n = 6$ map were audited adversarially, including a from-scratch symbolic re-derivation of the region of H_3 (same set and same minimal polynomial). The $n = 7$ prefilter and resolver were audited twice (979,800 exact comparisons; all 786,596 undecided lines recomputed independently, byte-identical), and the ray reduction, curve test and edge bound once (an independent all-minors computation on 3,129 graphs and 6,000 random exact legal points on never-stable graphs). The family theorems (Theorems 7.3, 7.5, 7.6, 7.7, 7.8), the 626-entry catalogue and the threshold tool were each reviewed by a separate reviewer internal to the project (not external peer review) who re-derived the steps, taking Lemma 2.4(b),(c) as given; no blocking issue was found. Non-fatal defects found by the referees (e.g. a misprinted node count, $2p(1) - 1$ for $2p(1) - 3$, in the notes on Theorem 7.7) are logged and corrected here.

8.3 Two machine campaigns at $n = 8$

The 9,133 eight-node limit counterexamples were verified twice by machine, with no human hand check. In the first campaign four verifiers were written from the definitions, without the project's code and each with its own digraph6 decoder: one with exact fractions, two characteristic-polynomial algorithms required to agree, and Hurwitz determinants cross-checked by a Routh array (45,665 points: all four stored witnesses and one own interior point per graph); one by exact right-half-plane root counting without Hurwitz determinants (Routh array and Cauchy index on 18,282 points, exact complex root isolation on every sixth graph); one checking structure (pairwise non-isomorphism by its own canonical form and nauty, non-completeness, domination-freeness, the stored interval data, and its own legal stable positive point for every graph, with the upper ends of all q-intervals refined); and one in a computer-algebra system (first legal witness on every graph, endpoint sharpness on every fiftieth, 183/183 sharp, and a Sturm count of one on every stored isolating interval). Every verifier confirmed all 9,133, with 0 refuted, 0 undecided, 0 vanishing Hurwitz determinants and 0 disagreements among them or with the 36,532 stored witness flags; controls (K_8, Geneson's graph, and 20 "never" graphs at 280 legal points) behaved as required.[46] The second campaign used six lenses the first had not: completeness from the seven-node side (858 closed-twin blow-ups of the 123 seven-node limit counterexamples, of which the 61 in the list are confirmed and the 797 others not limit counterexamples and not stable and positive at 4,782 sampled legal points; the 29,116 bidirected in-twin switches of listed graphs, all in the list with identical intervals); stability for *all* large t by exact root isolation in t (every graph: one interval (t^*, ∞)); the never side (600 "never" records at 874,996 legal points on their rays, 0 stable and positive, and an independent all-β replay reproducing every verdict; the prefilter's boundary rule re-derived on 36.9 million regenerated digraphs, 0 disagreements); a floating-point dynamics lens, explicitly non-decisional; an adversarial referee of the enumeration chain (edge bound, Burnside counts on 30 graphs, two shards and 20 dense graphs, coverage, agreement with the prefilter on 3,391 digraphs, accounting); and the hand-check kit. Its synthesis found 0 fatal and 0 major defects (one referee's MAJOR, an unfinished re-derivation of five never_full certificates, was covered by two other checks and reclassified), 13 explained non-issues and one minor documentation defect (the B lines had been described as "$\alpha = -1/2$ exactly"; they mean $\alpha \geq -1/2$, which does not affect their exclusion from the limit map but is why they cannot be reused for finite-ε questions).[47]

[46] `notes/verify_n8/SYNTHESIS.json` and the four directories under `notes/verify_n8/`.

[47] `notes/verify_n8_2/SYNTHESIS.json` and the six directories under `notes/verify_n8_2/`; `certificates/n8_limit_summary.json` (`machine_confirmation`, `second_verification`).

8.4 Human verification

The two new six-node counterexamples were hand-verified on 14 October 2026, and all 124 seven-node counterexamples on 15 October 2026; the project owner recorded the latter hand check as not fully confident and made sign-off conditional on the machine re-verification below. The 124 were additionally re-decided by four exact verifiers written from the definitions without access to the project's code: one with exact rationals and its own Hurwitz determinants; one with pure fractions and its own elimination and characteristic polynomials (with the transpose convention reversed, 0 of the 124 survive, a useful convention check); one by exact right-half-plane root counting without Hurwitz determinants (Routh array, Sturm sequences and exact isolation, required to agree); and one checking structural claims (pairwise non-isomorphism, domination-freeness, non-completeness, the exact limit test and $\alpha(E_0) = -1/2$). All 124 were confirmed at 742 legal points with 0 undecided cases.[48] The verification also found that 15 of the first stored witnesses in the limit file are illegal points ($q \leq 1$); a separate file lists one legal witness per graph.

No graph on eight or more nodes has been hand-verified. The project owner declined a hand check of the $n = 8$ list on 28 October 2026 and ordered the second machine campaign above instead; that decision is recorded in the certificate, and the $n = 8$ results are labelled "machine-verified" throughout to mark the difference in status from the $n \leq 7$ maps. A hand-check kit with six fully worked eight-node cases in exact fractions is provided for any reader who wishes to close this gap for a sample.[49]

8.5 Nondegeneracy

The conjecture is usually stated for nondegenerate CTLNs. At a witness point of each of the three six-node and 124 seven-node counterexamples we certified exactly that every one of the $2^n - 1$ supports has $\det(I - W_\sigma) \neq 0$, a fixed point with no zero coordinate and no off-support drive equal to 0, and that the full support is stable there. This check has not been run on the eight-node list.[50]

8.6 Machine-checked statements

Lean 4 (toolchain v4.35.0) with Mathlib (tag v4.35.0) checks 87 theorems with no `sorry`; the axiom audit shows only `propext`, `Classical.choice` and `Quot.sound`. They comprise: the closed form (4) with the transpose and $I - W = \varepsilon K$; the legality equivalence (5); Hurwitz stability of $-I + W$ iff positive stability of K; the ray identity, characteristic-polynomial scaling and propagation of stability down rays (Lemma 3.3); the intermediate-value step with the explicit root, and the forward direction of Lemma 3.4; Lemma 2.3; Lemma 3.6 including its spectral step; Lemma 3.5 with its induced-subgraph form; Stodola's condition (Lemma 3.7); and the twin eigenvalue (Lemma 3.8). Not formalised: the converse of Lemma 3.4 (it needs continuity of eigenvalues), the limit criterion Lemma 2.4, the rank-one identities of Lemma 2.2(ii)–(v), the off-support condition, and the full certificate lemma "(1)–(3) imply a stable equilibrium of (1) with support σ".[51]

Trust base. The certificates rely on: exact polynomial arithmetic of a standard computer-algebra system for the projection polynomials of the decomposition (determinants, factorisation, gcd, resultants, discriminants; the projection identities were re-verified by independent fraction-free determinants on a sample of 321 records); the Routh–Hurwitz theorem, which is cited; nauty's `geng`/`directg` for the seven-node enumeration, validated by its count, and for the restricted eight-node enumeration, validated by the exact Burnside count of every underlying graph, together with the two re-enumeration

[48] `notes/verify_124_workflow.json`; `certificates/n7_candidates_legal_witnesses.json`.

[49] `certificates/n8_limit_summary.json` (`status`, `second_verification.human_decision`); `notes/verify_n8_2/handcheck_kit/HANDCHECK.md`.

[50] `certificates/nondegeneracy_witnesses.json`, script `n7/nondegeneracy_check.py`.

[51] `notes/lean/README.md`, `notes/lean/CtlnCert/`.

schemes of Section 6, validated by canonical-form multiset equality at $n \le 7$ and by the $K_8 - e$ content check; and the C prefilters, validated by reproducing the six-node data exactly (the eight-node prefilter is byte-identical to the seven-node one on all six-node digraphs and on the 638,042 seven-node spectral survivors). The eight-node limit map also rests on the totals the prefilter reports for graphs with no survivor, which the accounting checks against the exact counts.

9 Open problems

Question 9.1 (Growth rate)**.** How fast does q_n^* grow? Theorem 7.7 gives $q_n^* \to \infty$ with a super-exponential $n_0(N)$; the exact values are 1.1664, 1.3247 and (in the limit) 1.7107 at $n = 6, 7, 8$, and the certified points are q close to 2 at $n = 19$, 5/2 at $n = 47$ and 23/5 for $n \ge 4{,}909$, while the only upper bound is Geneson's $n - 2$. Is $q_n^* \ge c \log n$, or $q_n^* \ge n^c$? The bottleneck is integer polynomials with $c_1 = 1$, nonnegative coefficients, roots in a thin strip $\operatorname{Re} z > -1/N$, and small value at 1.

Question 9.2 (The finite-ε map at $n = 8$)**.** The limit map of Section 6 is complete, but the finite-ε map at $n = 8$ is not: the 417,603,634 prefilter lines with $\alpha \ge -1/2$ were not stored, and finite-ε-only counterexamples on eight nodes exist (E_1 of Theorem 7.5 is one of the 15 graphs satisfying Lemma 7.4). At $n = 7$ the finite map was computed by the full chain of Table 2, and the bulge criterion (Theorem 5.4) accounts for its difference from the limit map; a re-run storing those lines, followed by the curve test of Section 3 on them and on the 7,078 never_full lines and the bulge rule on the 9,133, would settle whether $q_8^* = q_8^*(\infty)$ for all graphs with $\alpha \le -1/2$; the lines with $\alpha > -1/2$ need either a proof answering Question 9.5 or the full finite pipeline. The full finite-ε map needs the $n = 7$ pipeline over the 801,558,826,511 restricted digraphs; extrapolated from measured rates this is of the order of 4×10^4 CPU-hours. Is the union of the 9,133 limit intervals all of $(1, q_8^*(\infty))$?

Question 9.3 (Structure)**.** Prove or refute Conjecture 6.3. (S1) reduces, via the rank-4 structure of D in the presence of a bidirected K_{n-2}, to a finite-type family of integer matrices and holds exhaustively for $n \le 9$; (S2) is related to, but not implied by, Curto and Morrison's question whether proper sources die. Test (S2) and (S3) at $n = 9$ on the one-node extensions of the 9,133 and on the digraphs without an induced oriented 3-cycle.

Question 9.4 (Formalisation)**.** Formalise the converse of Lemma 3.4 (openness of positive stability), the limit criterion Lemma 2.4, and the certificate lemma (1)–(3)$\Rightarrow$ stable equilibrium, so that every certified case becomes a checked instance.

Question 9.5 (Characterising limit-stable graphs)**.** Proposition 7.9 refutes the natural spectral obstruction. Which graphs are limit-stable, i.e. have $w(\gamma) > 0$ and $\alpha < -\gamma$ at a common $\beta > 2$? A related question: is $\alpha \le -1/2$ for every graph whose full support is an SFP at some legal point? This holds for $n \le 7$ (Corollary 5.2), with equality attained for every $n \ge 7$ by Theorem 7.5; no proof is known.

Question 9.6 (Proper supports)**.** The truth of the conjecture depends only on full supports (Theorem 2.6), but listing all pairs (G, σ) realising a counterexample needs condition (2): describe the admissible in-sets for the $n \le 7$ counterexamples. At $n = 7$ all 12,288 pairs formed by a six-node counterexample and a seventh node with any in/out pattern have been certified (7,744 stable somewhere, 4,544 never, 0 escalations; `certificates/n7_proper_supports_all.jsonl`; not hand-verified); a structural description is open.

A Certificates

Paths are relative to the project archive. "Script" is the program that regenerates the file.

claim	certificate	script
Geneson's graph reproduced	certificates/stage0_geneson_2607.21396.json	test suite, block 8
$n = 5$ package, diff zero	certificates/stage1_n5_diff.json	stage1_n5_diff.py
engine validation	certificates/validate_fastregion_n4.json, _n5.json, region2d_v2_validation.json	validate_fastregion.py, region2d.py
$k \leq 4$	certificates/region2d_n2to4_all.json	region2d.py
Theorem 4.1	certificates/region2d_n5_shard0of1*, n5_summary.json	region2d.py
Theorem 4.2	certificates/n6_summary.json, region_n6_limit_*, region2d_n6_shard*, region_n6_eps*, COUNTEREXAMPLE_CANDIDATE_n6_idx*.json	fastregion.py, region2d.py
twin classes	certificates/n6_survivor_equivalence_classes.json	notes/prover_scripts_J/
Theorem 4.4	certificates/qstar/qstar_n6.json	n7/qstar.py
Table 2	certificates/n7_finite_summary.json, n7_curve_TOTAL.json, n7_region2d/	n7/run_n7_curve.py, run_n7_region2d.py, aggregate_n7.py
limit map $n = 7$	certificates/COUNTEREXAMPLE_CANDIDATES_n7_limit.jsonl, n7_limit_tally.json	n7/run_n7_limit.py
the 124 graphs	certificates/COUNTEREXAMPLE_CANDIDATES_n7_finite.jsonl, n7_candidates_legal_witnesses.json	n7/aggregate_n7.py, notes/verify_124/
E_0	certificates/COUNTEREXAMPLE_CANDIDATE_n7_finite_FNFL.json, qstar/qstar_fnfl.json	n7/write_finite_certificates.py, n7/qstar.py
Proposition 5.3	notes/lemma_M_refuted.md, §2	notes/prover_scripts_M/
Proposition 5.5	certificates/n7_bulge_Rstar_exact.json, n7_limit_hits_finite_eps_bulge_witnesses.json	notes/bulge_exact/
Theorem 5.6	certificates/qstar/qstar_n7_finite.json, qstar_n7limit.json	n7/qstar.py--sharpen
Theorems 7.3, 7.5, 7.6, Table 7	notes/research_wf3/theory/family_*_certificate.json, family_E_witnesses.json, top_n8.json, cert_q_910.json, cert_q_11.json, families_catalogue_n{6,7}.json	family_*_proof.py, certify_top.py, certify_q.py, run_family_prover.py, verify_all.py
eight-node instances	certificates/n8_instances_region2d.json, notes/research_wf3/theory/cube_n8_witnesses.json	n7/n8_instances_region2d.py, witness_cube.py
Theorem 7.7	certificates/theorem_U_explicit_instances.json, notes/theorem_U/	theorem_U.py, certify_forest.py, certify_explicit.py
Theorem 7.8	certificates/theorem_A_fixed_parameter_instances.json, notes/theorem_A_fixed/	theorem_A.py, independent_A.py
nondegeneracy	certificates/nondegeneracy_witnesses.json	n7/nondegeneracy_check.py
Table 3, Theorem 6.1	certificates/n8_limit_summary.json, n8_summary.json, n8/n8_prefilter_TOTAL.txt, n8_accounting.json, n8_restricted_counts.json, n8_limit_tally.json, n8/limit_sharding/shard*_tally.json	n7/filter8.c, n8/check_accounting.py, n8/run_n8_limit.py, n8/aggregate_n8.py; runbook n7/REPRODUCE_n8_limit.md
enumeration schemes at $n = 8$	notes/k8_decomposition.md, k8_decomposition_referees.json, n8/check_k8decomp.log, n8/check_orbit.log, n8/orbitenum/k8e_check/	n8/k8decomp/, n8/orbitenum/
the 9,133 graphs	n8/n8_limit_HITS.jsonl, certificates/n8_candidates_legal_witnesses.json, n8_counterexamples_classes.json	n8/aggregate_n8.py
Theorem 6.2	certificates/n8_summary.json (q_star_limit, q_data), notes/verify_n8/v_structure/q_upper_ends_refined.json	n8/aggregate_n8.py
Section 6.5, Conjecture 6.3	notes/n8_structure.md, notes/n8_structure_tools/*.json	notes/n8_structure_tools/*.py

claim	certificate	script
$n = 8$ machine verification	`notes/verify_n8/SYNTHESIS.json`, `notes/verify_n8_2/SYNTHESIS.json`, `notes/verify_n8_2/handcheck_kit/HANDCHECK.md`	`notes/verify_n8/`, `notes/verify_n8_2/`
Lean	`notes/lean/CtlnCert/`	`lake build`, `Audit.lean`

Acknowledgments

The enumeration, certification and verification pipelines described in Section 8 were built and run by automated software agents under the author's direction; every statement in this paper rests on the exact, re-runnable certificates in the accompanying repository, and the author is responsible for the content. The $n = 5$ reproduction uses the published package of Curto and collaborators; nauty (geng, directg, labelg) was used for graph enumeration and canonical labelling.

References

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